Solutions to the Exercises of Chapter 7 (Limit Cycles)

Strogatz, *Nonlinear Dynamics and Chaos* (2nd ed.)

Conventions. Dots denote $\mathrm{d}/\mathrm{d}t$ (primes in Section 7.6 denote $\mathrm{d}/\mathrm{d}\tau$ or $\mathrm{d}/\mathrm{d}T$, as stated). Polar coordinates are $x = r\cos\theta$, $y = r\sin\theta$, and I use the identities

$$r\dot{r} = x\dot{x} + y\dot{y}, \qquad r^2\dot{\theta} = x\dot{y} - y\dot{x}.$$

Problems that ask for a "sketch" are answered with a description of the phase portrait (fixed points, their types, invariant curves, flow direction); a few are accompanied by a figure. Problems that require a computer were done numerically (Python/SciPy, `solve_ivp` with tolerances 10^{-10} or better) and the numbers are reported. Where the printed statement of a problem seems to contain a typo or an undefined symbol, I say so explicitly.

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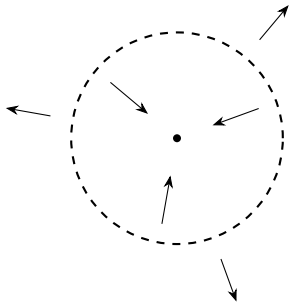
6 Weakly Nonlinear Oscillators (7.6) **24**

1 Examples (7.1)

7.1.1 *Sketch the phase portrait of* $\dot{r} = r^3 - 4r$, $\dot{\theta} = 1$.

Solution. Here $\dot{r} = r(r-2)(r+2)$, so for $r \geq 0$ the only radial equilibria are $r = 0$ and $r = 2$. For $0 < r < 2$ we have $\dot{r} < 0$; for $r > 2$ we have $\dot{r} > 0$. Since $\dot{\theta} = 1$ everything rotates counterclockwise at unit angular speed.

- The origin is the only fixed point; linearization gives $\dot{r} \approx -4r$, so it is a *stable spiral* (counterclockwise).
- $r = 2$ is a closed orbit (period 2π) and it is *unstable*: trajectories starting inside spiral into the origin, those starting outside spiral off to infinity.

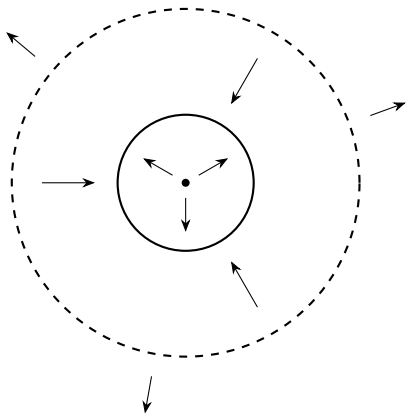


origin stable; dashed circle $r = 2$ unstable

■

7.1.2 *Sketch the phase portrait of* $\dot{r} = r(1 - r^2)(9 - r^2)$, $\dot{\theta} = 1$.

Solution. The radial equilibria are $r = 0, 1, 3$. Signs of $\dot{r}$: positive on $(0, 1)$, negative on $(1, 3)$, positive on $(3, \infty)$. The origin is an *unstable spiral* ($\dot{r} \approx 9r$), $r = 1$ is a *stable* limit cycle, $r = 3$ is an *unstable* limit cycle, and trajectories outside $r = 3$ spiral off to infinity (all counterclockwise).



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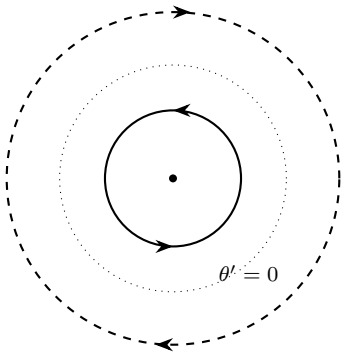
7.1.3 *Sketch the phase portrait of* $\dot{r} = r(1 - r^2)(4 - r^2)$, $\dot{\theta} = 2 - r^2$.

Solution. Radial equilibria: $r = 0, 1, 2$; $\dot{r} > 0$ on $(0, 1)$, $\dot{r} < 0$ on $(1, 2)$, $\dot{r} > 0$ for $r > 2$. So the origin is an unstable spiral, $r = 1$ is a stable limit cycle, $r = 2$ is an unstable limit cycle.

The new feature is the angular motion $\dot{\theta} = 2 - r^2$: it is positive for $r < \sqrt{2}$ (counterclockwise), zero on $r = \sqrt{2}$, and negative for $r > \sqrt{2}$ (clockwise). Hence

- the stable cycle $r = 1$ ($\dot\theta = 1$) is traversed counterclockwise, with period 2π;
- the unstable cycle $r = 2$ ($\dot\theta = -2$) is traversed *clockwise*, with period π;
- on the circle $r = \sqrt{2} \in (1, 2)$ the motion is purely radial (inward, since $\dot r < 0$ there): trajectories between the two cycles spiral counterclockwise while $1 < r < \sqrt{2}$, cross the circle $r = \sqrt{2}$ moving straight inward, then spiral clockwise for $\sqrt{2} < r < 2$, ... and in the region $1 < r < 2$ they all end up on $r = 1$;
- outside $r = 2$, trajectories spiral clockwise out to infinity.

The origin is the only fixed point.



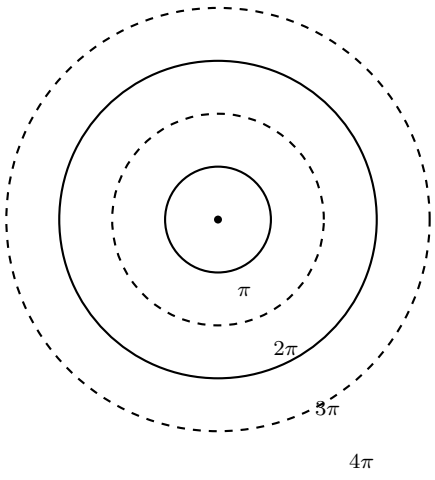
(solid: stable cycle, counterclockwise; dashed: unstable cycle, clockwise; dotted: $r = \sqrt{2}$.) ■

7.1.4 *Sketch the phase portrait of* $\dot r = r \sin r$, $\dot\theta = 1$.

Solution. $\dot r = 0$ at $r = n\pi$, $n = 0, 1, 2, \dots$, and $\operatorname{sign} \dot r = \operatorname{sign} \sin r$, which is $+$ on $(0, \pi)$, $-$ on $(\pi, 2\pi)$, $+$ on $(2\pi, 3\pi)$, and so on, alternating.

- $r = 0$: $\dot r \approx r^2 > 0$, so the origin is a (non-hyperbolic) *unstable* spiral, trajectories leave it counterclockwise.
- $r = (2k+1)\pi$ ($k = 0, 1, 2, \dots$): $\dot r$ changes from $+$ to $-$, so these closed orbits are *stable* limit cycles.
- $r = 2k\pi$ ($k = 1, 2, \dots$): $\dot r$ changes from $-$ to $+$, so these are *unstable* limit cycles.

Thus there are infinitely many concentric limit cycles of period 2π, alternately stable (radii $\pi, 3\pi, 5\pi, \dots$) and unstable (radii $2\pi, 4\pi, \dots$), and every trajectory other than the origin is attracted to a stable cycle or lies on an unstable one.



■

7.1.5 *Show that* $\dot r = r(1-r^2)$, $\dot\theta = 1$ *is equivalent to* $\dot x = x - y - x(x^2+y^2)$, $\dot y = x + y - y(x^2+y^2)$.

Solution. With $x = r\cos\theta$, $y = r\sin\theta$,

$$\dot x = \dot r\cos\theta - r\dot\theta\sin\theta = r(1-r^2)\cos\theta - r\sin\theta = x(1-r^2) - y = x - y - x(x^2+y^2),$$

$$\dot y = \dot r\sin\theta + r\dot\theta\cos\theta = r(1-r^2)\sin\theta + r\cos\theta = y(1-r^2) + x = x + y - y(x^2+y^2).$$

Conversely the transformation is invertible for $r > 0$, so the two systems are equivalent. ■

7.1.6 *(Circuit for van der Pol oscillator.) (a) Let* $V = V_{32} = -V_{23}$. *Show* $\dot V = -I/C$ *and* $V = L\dot I + f(I)$. *(b) Show these are equivalent to* $\mathrm{d}w/\mathrm{d}\tau = -x$, $\mathrm{d}x/\mathrm{d}\tau = w - \mu F(x)$ *where* $x = L^{1/2}I$, $w = C^{1/2}V$, $\tau = (LC)^{-1/2}t$, $F(x) = f(L^{-1/2}x)$.

Solution. *(a)* Take I to be the current flowing $1 \to 2$ through the inductor, then from 2 to 3 through the capacitor, then $3 \to 4$ through the nonlinear element, and $4 \to 1$ along the wire. For the capacitor, $C\,\mathrm{d}(V_2 - V_3)/\mathrm{d}t = I$; since $V = V_{32} = V_3 - V_2$, this says

$$C\dot V = -I \quad\Longrightarrow\quad \dot V = -I/C.$$

For the voltage drops around the loop, $V_3 - V_2 = (V_3 - V_4) + (V_4 - V_1) + (V_1 - V_2)$. The element gives $V_3 - V_4 = f(I)$, the wire gives $V_4 - V_1 = 0$, and the inductor gives $V_1 - V_2 = L\dot I$. Hence

$$V = L\dot I + f(I).$$

(b) Put $x = L^{1/2}I$, $w = C^{1/2}V$, $\tau = (LC)^{-1/2}t$, so $\mathrm{d}t/\mathrm{d}\tau = (LC)^{1/2}$. Then

$$\frac{\mathrm{d}w}{\mathrm{d}\tau} = C^{1/2}\dot V\,(LC)^{1/2} = C^{1/2}\Big(-\frac{I}{C}\Big)(LC)^{1/2} = -L^{1/2}I = -x,$$

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = L^{1/2}\dot I\,(LC)^{1/2} = L^{1/2}(LC)^{1/2}\,\frac{V - f(I)}{L} = C^{1/2}\big(V - f(I)\big) = w - C^{1/2}F(x),$$

because $f(I) = f(L^{-1/2}x) = F(x)$. This is the stated form with $\mu = C^{1/2}$. (*Remark:* the printed statement does not define μ; the computation shows $\mu = C^{1/2}$. One can also absorb it by rescaling F.) ■

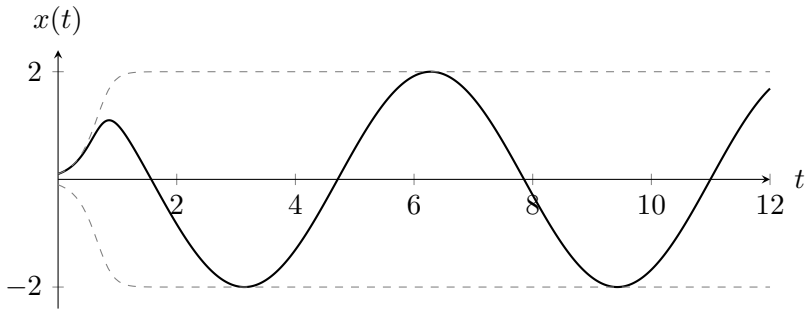
7.1.7 *(Waveform) For* $\dot r = r(4-r^2)$, $\dot\theta = 1$, $x(t) = r(t)\cos\theta(t)$, $x(0) = 0.1$, $y(0) = 0$, *sketch* $x(t)$ *without an explicit expression.*

Solution. The initial point has $r(0) = 0.1$, $\theta(0) = 0$, and $\theta(t) = t$. The radial equation has a stable fixed point at $r = 2$ and an unstable one at $r = 0$; since $r(0) = 0.1 > 0$ small, $r(t)$ grows from 0.1 and saturates at 2. Near the origin $\dot r \approx 4r$, so $r \approx 0.1e^{4t}$ initially (a fast exponential growth, time scale $1/4$), and it levels off at 2 after $t = O(1)$. Therefore

$$x(t) = r(t)\cos t \quad \text{is a cosine of period } 2\pi \text{ whose envelope } r(t) \text{ rises quickly from 0.1 to 2.}$$

Qualitatively: x starts at 0.1, climbs to a first positive hump of height only about 1 (the envelope is still growing during the first quarter-period, and $\cos t$ is already falling), crosses zero near $t = \pi/2$, and from then on is essentially $2\cos t$: a steady oscillation of amplitude 2 and period 2π.

(*Check, not needed for the sketch:* with $u = r^2$ the radial equation becomes logistic, $\dot u = 2u(4-u)$, giving $r(t) = 2/\sqrt{1 + 399e^{-8t}}$. Numerically $x(0.5) = 0.61$, $x(1) = 1.01$, $x(1.5) = 0.14$, $x(2) = -0.83$, $x(3) = -1.98$.)



■

7.1.8 *(A circular limit cycle) Consider $\ddot{x} + a\dot{x}(x^2 + \dot{x}^2 - 1) + x = 0$, $a > 0$. (a) Find and classify all fixed points. (b) Show there is a circular limit cycle; find amplitude and period. (c) Determine its stability. (d) Argue that it is unique.*

Solution. Write $y = \dot{x}$, so $\dot{x} = y$, $\dot{y} = -x - ay(x^2 + y^2 - 1)$.

(a) Fixed points: $y = 0$ and then $x = 0$. So the origin is the only fixed point. The Jacobian there is $\begin{pmatrix} 0 & 1 \\ -1 & a \end{pmatrix}$ (because $-ay(r^2 - 1) \to +ay$), with $\tau = a > 0$, $\Delta = 1 > 0$, $\tau^2 - 4\Delta = a^2 - 4$. Hence the origin is an *unstable spiral* for $0 < a < 2$ and an *unstable node* for $a \geq 2$ (a degenerate node at $a = 2$).

(b) In polar coordinates,

$$r\dot{r} = xy + y\big(-x - ay(r^2 - 1)\big) = -ay^2(r^2 - 1) \implies \dot{r} = a\, r \sin^2\theta\,(1 - r^2),$$

$$r^2\dot{\theta} = x\dot{y} - y\dot{x} = -x^2 - axy(r^2 - 1) - y^2 \implies \dot{\theta} = -1 - a \sin\theta\cos\theta\,(r^2 - 1).$$

On $r = 1$ we get $\dot{r} = 0$ and $\dot{\theta} = -1$. So the unit circle is a closed orbit, traversed *clockwise* with $x(t) = \cos(t + \text{const})$: amplitude 1 and period 2π.

(c) Let $\rho = r - 1$. Linearizing $\dot{r}$ about $r = 1$: $\dot{\rho} \approx -2a \sin^2\theta\, \rho$, and on the cycle $\theta = -t + \text{const}$. Over one period the Floquet multiplier is

$$\exp\Big(-2a \int_0^{2\pi} \sin^2 t \, \mathrm{d}t\Big) = e^{-2\pi a} < 1.$$

Hence the limit cycle is *stable*. (Equivalently: $\dot{r} \geq 0$ for $r < 1$ and $\dot{r} \leq 0$ for $r > 1$.)

(d) Uniqueness. The circle $r = 1$ is itself a trajectory, so no other trajectory can cross it. Suppose there were a closed orbit with $r < 1$ throughout. Then $\dot{r} = ar \sin^2\theta(1 - r^2) \geq 0$ with equality only when $\sin\theta = 0$, i.e. at isolated instants (at $y = 0$, $x \neq 0$ we have $\dot{y} = -x \neq 0$, so the orbit does not stay on the x-axis). Hence r would strictly increase over a period, which is impossible for a closed orbit. The same argument with $\dot{r} \leq 0$ rules out closed orbits with $r > 1$. A closed orbit cannot pass through the origin (a fixed point). Therefore $r = 1$ is the only periodic orbit. ■

7.1.9 *(Circular pursuit) A dog starts at the centre of a unit circular pond; a duck swims on the edge counterclockwise at unit speed; the dog swims directly toward the duck. (a) For equal speeds derive and analyze the equations for the dog's path. Can you solve them? Does the dog catch the duck? (b) Dog k times faster: derive the equations. (c) If $k = \frac{1}{2}$, what does the dog do in the long run?*

Solution. Let the dog be at polar position (R,θ) and the duck at $(1,\phi)$. The duck moves at unit speed on the unit circle, so $\dot\phi = 1$ (we may take $\phi = t$). The vector from dog to duck is

$$\mathbf{d} = (\cos\phi - R\cos\theta,\ \sin\phi - R\sin\theta), \qquad D = |\mathbf{d}| = \sqrt{1 + R^2 - 2R\cos(\phi-\theta)}.$$

If the dog's speed is s (so $s = 1$ in (a), $s = k$ in (b), with the duck's speed fixed at 1), its velocity is $s\,\mathbf{d}/D$. Projecting on the radial and angular unit vectors $\mathbf{e}_r = (\cos\theta, \sin\theta)$, $\mathbf{e}_\theta = (-\sin\theta, \cos\theta)$ gives $\mathbf{d}\cdot\mathbf{e}_r = \cos(\phi-\theta) - R$ and $\mathbf{d}\cdot\mathbf{e}_\theta = \sin(\phi-\theta)$. Hence

$$\dot R = \frac{s\,[\cos(\phi-\theta) - R]}{D}, \qquad R\dot\theta = \frac{s\,\sin(\phi-\theta)}{D}, \qquad \dot\phi = 1.$$

Using ϕ as the independent variable (since $\phi = t$) these are the equations

$$\frac{\mathrm{d}R}{\mathrm{d}\phi} = \frac{s\,[\cos(\phi-\theta) - R]}{D}, \qquad \frac{\mathrm{d}\theta}{\mathrm{d}\phi} = \frac{s\,\sin(\phi-\theta)}{R\,D}.$$

Let $\psi = \phi - \theta$ be the angular lag of the dog behind the duck. Then the system closes in (R,ψ):

$$\dot R = \frac{s(\cos\psi - R)}{D}, \qquad \dot\psi = 1 - \frac{s\sin\psi}{R\,D}, \qquad D = \sqrt{1 + R^2 - 2R\cos\psi}. \tag{1}$$

(b) is exactly (1) with $s = k$.

(a), $s = 1$. Any steady circling requires $\dot R = 0$, $\dot\psi = 0$, i.e. $R = \cos\psi$ and $\sin\psi = RD$. With $R = \cos\psi$ one finds $D^2 = 1 - R^2 = \sin^2\psi$, so the second condition reads $\sin\psi = \cos\psi\sin\psi$, i.e. $\cos\psi = 1$: this is $R = 1, \psi = 0$, $D = 0$, the capture configuration where the equations are singular. There is no regular steady state, and no closed-form solution of the dog's path is known.

What can be shown: the distance obeys $\dot D = \cos\alpha - 1 \le 0$, where α is the angle between the line of sight and the duck's velocity, so D is nonincreasing. On the pond's edge $R = 1$ (with $\psi \neq 0$) we have $\dot R = (\cos\psi - 1)/D < 0$, i.e. the dog is pushed back inside, so the dog never reaches the edge except at the duck itself. Numerically (starting the dog at the centre), $R \to 1$ and $D \to 0$ like $D \approx 2/t$:

t	10	30	60	100	200
R	0.9940	0.9987	0.9996	0.9998	0.99995
D	0.108	0.0518	0.0291	0.0184	0.0096

So the dog gets arbitrarily close to the duck, trailing it just inside the edge, but *never catches it in finite time.*

(c), $s = k = \frac12$. Now the steady state exists: $R = \cos\psi$ and $k\sin\psi = RD = \cos\psi\sin\psi$ give $\cos\psi = k$. So $R^* = k = \frac12$, $\psi^* = \arccos k = \pi/3$, $D^* = \sin\psi^* = \sqrt3/2$. The Jacobian of (1) at this point is (numerically evaluated, with entries $-1/\sqrt3,\ -1/2,\ 2,\ 0$) $\begin{pmatrix} -0.577 & -0.5 \\ 2 & 0 \end{pmatrix}$, with eigenvalues $-0.289 \pm 0.957\,i$: a *stable spiral.* So in the long run the dog swims in a circle of radius $R = k = \frac12$ about the pond's centre with the same angular speed as the duck, staying 60° behind it (its linear speed is $R\dot\theta = \frac12 = k$, consistent). It never catches the duck. A numerical integration confirms convergence to $R = 0.50000$, $\psi = 1.0472$ by $t \approx 60$. ■

2 Ruling Out Closed Orbits (7.2)

7.2.1 *Plot the phase portrait of the gradient system $\dot{\mathbf{x}} = -\nabla V$ for $V = x^2 + y^2$.*

Solution. $\dot{x} = -2x$, $\dot{y} = -2y$. The origin is a *stable star* (proper node): all trajectories are straight rays $y = cx$ (and the axes) running into the origin; $V = r^2$ decreases along them, and the equipotentials are circles centred at the origin, which the rays cross at right angles. Solutions: $(x, y) = (x_0, y_0)e^{-2t}$. ■

7.2.2 *Same for $V = x^2 - y^2$.*

Solution. $\dot{x} = -2x$, $\dot{y} = +2y$: a *saddle* at the origin with stable manifold the x-axis and unstable manifold the y-axis. Since xy is constant ($\frac{\mathrm{d}}{\mathrm{d}t}(xy) = -2xy + 2xy = 0$), trajectories are the hyperbolas $xy = c$, flowing in along the x-direction and out along the y-direction. The equipotentials $x^2 - y^2 = c$ are the orthogonal family of rotated hyperbolas (with $V = 0$ on the lines $y = \pm x$). ■

7.2.3 *Same for $V = e^x \sin y$.*

Solution. $\dot{x} = -e^x \sin y$, $\dot{y} = -e^x \cos y$. Since $e^x > 0$, a fixed point would need $\sin y = \cos y = 0$, which is impossible, so there are *no fixed points*. Dividing, $\mathrm{d}y/\mathrm{d}x = \cot y$, i.e. $\tan y \, \mathrm{d}y = \mathrm{d}x$, so trajectories are the curves $\cos y = Ce^{-x}$ (while the equipotentials $e^x \sin y = c$ are the orthogonal family). The lines $y = -\pi/2 + 2n\pi$ are invariant ($\dot{y} = 0$ there, and $\dot{x} = +e^x > 0$): the flow runs to the right ($x \to +\infty$) along them, and they are *attracting* in the y-direction (near $y = -\pi/2$, $\dot{y} \approx -e^x(y + \pi/2)$). The lines $y = \pi/2 + 2n\pi$ are invariant with flow to the left ($\dot{x} = -e^x$) and are *repelling* in y. For $C > 0$ the trajectory $\cos y = Ce^{-x}$ is a hairpin in the strip $|y| < \pi/2$, open to the right and with its vertex at $(\ln C, 0)$: the upper half comes in from $x = +\infty$ near the repelling line $y = \pi/2$ moving left and down, turns around at the vertex, and the lower half runs off to the right toward the attracting line $y = -\pi/2$. Each strip $-\pi/2 + 2n\pi < y < \pi/2 + 2n\pi$ is filled with such hairpins, and there is a second family (with $C < 0$) in the strips centred on $y = \pi$. ■

7.2.4 *Show that all vector fields on the line are gradient systems. Is the same true for vector fields on the circle?*

Solution. On the line, $\dot{x} = f(x)$. Define $V(x) = -\int_0^x f(s)\,\mathrm{d}s$; then $-V'(x) = f(x)$, so $\dot{x} = -V'(x)$ is a gradient system.

On the circle, $\dot{\theta} = f(\theta)$ with f 2π-periodic. A potential $V(\theta)$ with $\dot{\theta} = -V'(\theta)$ must be a single-valued (2π-periodic) function on the circle, which requires $V(2\pi) = V(0)$, i.e.

$$\int_0^{2\pi} f(\theta)\,\mathrm{d}\theta = 0.$$

So *not* every vector field on the circle is a gradient system: e.g. $\dot{\theta} = 1$ (uniform rotation) is not, since a gradient flow can never go around a closed orbit. A field on the circle is a gradient field exactly when $\int_0^{2\pi} f\,\mathrm{d}\theta = 0$, for instance $\dot{\theta} = \sin\theta$. ■

7.2.5 *Let $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ be a smooth vector field on the plane. (a) Show that if it is a gradient system, then $\partial f/\partial y = \partial g/\partial x$. (b) Is the condition sufficient?*

Solution. *(a)* If $f = -V_x$ and $g = -V_y$, then $f_y = -V_{xy} = -V_{yx} = g_x$ (equality of mixed partials for the smooth V).

(b) Yes, on the whole plane (or any simply connected domain). Suppose $f_y = g_x$ on $\mathbb{R}^2$ and define

$$V(x, y) = -\int_0^x f(s, 0)\,\mathrm{d}s - \int_0^y g(x, t)\,\mathrm{d}t.$$

Then $V_y = -g(x, y)$, and

$$V_x = -f(x,0) - \int_0^y g_x(x,t)\,\mathrm{d}t = -f(x,0) - \int_0^y f_y(x,t)\,\mathrm{d}t = -f(x,0) - \big(f(x,y) - f(x,0)\big) = -f(x,y).$$

So $(f, g) = -\nabla V$. (This is the Poincaré lemma: a closed 1-form $f\,dx + g\,dy$ on a simply connected domain is exact.) The hypothesis "simply connected" matters: on $\mathbb{R}^2 \setminus \{0\}$ the field $\dot{x} = -y/r^2$, $\dot{y} = x/r^2$ satisfies $f_y = g_x$, yet it has closed orbits and hence is not a gradient system. ■

7.2.6 *Find the potential V for (a) $\dot{x} = y^2 + y\cos x$, $\dot{y} = 2xy + \sin x$; (b) $\dot{x} = 3x^2 - 1 - e^{2y}$, $\dot{y} = -2xe^{2y}$.*

Solution. *(a)* Check: $f_y = 2y + \cos x = g_x$. From $V_x = -f = -y^2 - y\cos x$ we integrate in x: $V = -xy^2 - y\sin x + h(y)$. Then $V_y = -2xy - \sin x + h'(y)$ must equal $-g = -2xy - \sin x$, so $h' = 0$. Thus

$$V(x, y) = -xy^2 - y\sin x \quad (+\text{const}).$$

(b) Check: $f_y = -2e^{2y} = g_x$. From $V_x = -3x^2 + 1 + e^{2y}$: $V = -x^3 + x + xe^{2y} + h(y)$; then $V_y = 2xe^{2y} + h'(y)$ must equal $-g = 2xe^{2y}$, so $h' = 0$:

$$V(x, y) = -x^3 + x + xe^{2y} \quad (+\text{const}).$$

■

7.2.7 *Consider $\dot{x} = y + 2xy$, $\dot{y} = x + x^2 - y^2$. (a) Show $f_y = g_x$. (b) Find V. (c) Sketch the phase portrait.*

Solution. *(a)* $f_y = 1 + 2x$ and $g_x = 1 + 2x$.

(b) $V_x = -y - 2xy \Rightarrow V = -xy - x^2y + h(y)$; $V_y = -x - x^2 + h'(y) = -g = -x - x^2 + y^2$ gives $h' = y^2$, so

$$V(x, y) = -xy - x^2y + \tfrac{1}{3}y^3.$$

(c) Fixed points: $f = y(1 + 2x) = 0$ gives $y = 0$ or $x = -\frac{1}{2}$. If $y = 0$: $x + x^2 = 0$, so $x = 0$ or -1. If $x = -\frac{1}{2}$: $g = -\frac{1}{4} - y^2 \neq 0$. So the fixed points are $(0, 0)$ and $(-1, 0)$. The Jacobian is $\begin{pmatrix} 2y & 1+2x \\ 1+2x & -2y \end{pmatrix}$: at $(0,0)$ it is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (eigenvalues ± 1, a saddle with unstable direction $(1, 1)$, stable direction $(1, -1)$); at $(-1, 0)$ it is $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ (eigenvalues ± 1, a saddle with unstable direction $(1, -1)$, stable direction $(1, 1)$).

So the portrait has two saddles on the x-axis, no closed orbits, and V decreasing along trajectories. Both saddles lie on the level set $V = 0$, which consists of the line $y = 0$ and the hyperbola $y^2 = 3x(1 + x)$. There are no sinks: since $V \sim \frac{1}{3}y^3$ for large $|y|$, every trajectory other than the stable manifolds of the two saddles runs off to infinity (numerically, in finite time), either toward $x \to \pm\infty$ with y large and positive, or downward along the vertical line $x = -\frac{1}{2}$ toward $y \to -\infty$ (this line is the axis of symmetry of the picture, since $\dot{x} \propto 1 + 2x$ vanishes there). The unstable manifolds of the saddles lead to the downward escape; the four stable manifolds separate the different escape regions. ■

7.2.8 *Show that trajectories of a gradient system always cross the equipotentials at right angles (except at fixed points).*

Solution. Let $\dot{\mathbf{x}} = -\nabla V$. At a point that is not a fixed point, $\nabla V \neq 0$, so the level set $V = c$ through that point is a smooth curve. If $\mathbf{t}$ is a tangent vector to it, then differentiating $V = c$ along the curve gives $\nabla V \cdot \mathbf{t} = 0$. The trajectory's tangent is $\dot{\mathbf{x}} = -\nabla V$, so $\dot{\mathbf{x}} \cdot \mathbf{t} = -\nabla V \cdot \mathbf{t} = 0$. Hence trajectories are orthogonal to the equipotentials. ■

7.2.9 *Decide whether each system is a gradient system; if so, find V, sketch the phase portrait and the equipotentials. (a) $\dot{x} = y + x^2y$, $\dot{y} = -x + 2xy$; (b) $\dot{x} = 2x$, $\dot{y} = 8y$; (c) $\dot{x} = -2xe^{x^2+y^2}$, $\dot{y} = -2ye^{x^2+y^2}$.*

Solution. *(a)* $f_y = 1 + x^2$ but $g_x = -1 + 2y$. These are not equal, so it is *not* a gradient system.

(b) $f_y = 0 = g_x$, so it is a gradient system with $V_x = -2x$, $V_y = -8y$:

$$V = -x^2 - 4y^2.$$

The origin is an *unstable node* (eigenvalues 2 and 8). Eliminating t, $\mathrm{d}y/\mathrm{d}x = 4y/x$ gives trajectories $y = Cx^4$ (plus the two axes): near the origin they are tangent to the slow direction, the x-axis, and they leave the origin more steeply the farther they go. The equipotentials are the ellipses $x^2 + 4y^2 = c$.

(c) $f_y = -4xye^{x^2+y^2} = g_x$, so it is a gradient system, and $V_x = 2xe^{r^2}$, $V_y = 2ye^{r^2}$ give

$$V = e^{x^2+y^2}.$$

In polar form $\dot{r} = -2re^{r^2}$, $\dot{\theta} = 0$: the origin is a *stable star*, trajectories are straight rays into the origin, with speed growing rapidly with r (in fact reaching the origin from large r very fast). Equipotentials are circles $r =$const. ■

7.2.10 *Show that $\dot{x} = y - x^3$, $\dot{y} = -x - y^3$ has no closed orbits by constructing a Liapunov function $V = ax^2 + by^2$.*

Solution. For $V = ax^2 + by^2$,

$$\dot{V} = 2ax(y - x^3) + 2by(-x - y^3) = 2(a - b)xy - 2ax^4 - 2by^4.$$

Choose $a = b = 1$: $V = x^2 + y^2$ and $\dot{V} = -2(x^4 + y^4) < 0$ for $(x, y) \neq (0, 0)$. So V is positive definite and strictly decreasing along every non-constant trajectory. On a closed orbit, V would return to its initial value after one period, but it is strictly decreasing, a contradiction. Hence there are no closed orbits. (In fact the origin is globally asymptotically stable.) ■

7.2.11 *Show that $V = ax^2 + 2bxy + cy^2$ is positive definite iff $a > 0$ and $ac - b^2 > 0$.*

Solution. ($\Leftarrow$) If $a > 0$, complete the square:

$$V = a\Big(x + \frac{b}{a}y\Big)^2 + \frac{ac - b^2}{a}\,y^2.$$

If $ac - b^2 > 0$, both coefficients are positive, so $V \geq 0$ and $V = 0$ forces $y = 0$ and then $x = 0$.

($\Rightarrow$) If V is positive definite, then $a = V(1, 0) > 0$. Take $(x, y) = (-b/a,\, 1)$: then $V = \frac{ac-b^2}{a} > 0$, hence $ac - b^2 > 0$ (since $a > 0$). ■

7.2.12 *Show that $\dot{x} = -x + 2y^3 - 2y^4$, $\dot{y} = -x - y + xy$ has no periodic solutions. (Hint: $V = x^m + ay^n$.)*

Solution. Try $V = x^2 + ay^4$. Then

$$\begin{aligned}\dot{V} &= 2x(-x + 2y^3 - 2y^4) + 4ay^3(-x - y + xy)\\ &= -2x^2 + 4xy^3 - 4xy^4 - 4axy^3 - 4ay^4 + 4axy^4.\end{aligned}$$

With $a = 1$ the cross terms cancel: $\dot{V} = -2x^2 - 4y^4$. So $m = 2$, $a = 1$, $n = 4$: $V = x^2 + y^4$ is positive definite and $\dot{V} < 0$ except at the origin. As in 7.2.10, a periodic orbit would force V to return to its initial value, contradicting strict decrease. Hence there are no periodic solutions. ■

7.2.13 *Using Dulac's criterion with $g = (N_1N_2)^{-1}$, show the competition model $\dot{N}_1 = r_1N_1(1 - N_1/K_1) - b_1N_1N_2$, $\dot{N}_2 = r_2N_2(1 - N_2/K_2) - b_2N_1N_2$ has no periodic orbits in $N_1, N_2 > 0$.*

Solution. (Here $r_i, K_i, b_i > 0$.) With $g = 1/(N_1N_2)$:

$$g\dot{N}_1 = \frac{r_1}{N_2}\Big(1 - \frac{N_1}{K_1}\Big) - b_1, \qquad g\dot{N}_2 = \frac{r_2}{N_1}\Big(1 - \frac{N_2}{K_2}\Big) - b_2.$$

Therefore

$$\nabla\cdot(g\dot{\mathbf{N}}) = \frac{\partial}{\partial N_1}(g\dot{N}_1) + \frac{\partial}{\partial N_2}(g\dot{N}_2) = -\frac{r_1}{K_1N_2} - \frac{r_2}{K_2N_1} < 0$$

throughout the open first quadrant. This region is simply connected and g is smooth there, so by Dulac's criterion there are no periodic orbits in it. ■

7.2.14 *Consider $\dot{x} = x^2 - y - 1$, $\dot{y} = y(x - 2)$. (a) Show there are three fixed points and classify them. (b) By considering the three straight lines through pairs of fixed points, show there are no closed orbits. (c) Sketch the phase portrait.*

Solution. *(a)* $\dot{y} = 0 \Rightarrow y = 0$ or $x = 2$. If $y = 0$: $x^2 = 1$, so $(\pm1, 0)$. If $x = 2$: $4 - y - 1 = 0$, so $y = 3$: $(2, 3)$. The Jacobian is $\begin{pmatrix} 2x & -1 \\ y & x - 2 \end{pmatrix}$.

- $(1, 0)$: $\begin{pmatrix} 2 & -1 \\ 0 & -1 \end{pmatrix}$, eigenvalues $2, -1$: *saddle.*
- $(-1, 0)$: $\begin{pmatrix} -2 & -1 \\ 0 & -3 \end{pmatrix}$, eigenvalues $-2, -3$: *stable node.*
- $(2, 3)$: $\begin{pmatrix} 4 & -1 \\ 3 & 0 \end{pmatrix}$, $\tau = 4$, $\Delta = 3$, eigenvalues $1, 3$: *unstable node.*

(b) The three lines are $y = 0$ (through $(\pm1, 0)$), $y = 3(x-1)$ (through $(1, 0)$ and $(2, 3)$) and $y = x+1$ (through $(-1, 0)$ and $(2, 3)$). Each is invariant:

- on $y = 0$: $\dot{y} = 0$;
- on $y = 3(x - 1)$: $\dot{x} = x^2 - 3x + 3 - 1 = (x - 1)(x - 2)$ and $\dot{y} = 3(x - 1)(x - 2) = 3\dot{x}$, so the field is tangent to the line;
- on $y = x + 1$: $\dot{x} = x^2 - x - 2 = (x - 2)(x + 1)$ and $\dot{y} = (x + 1)(x - 2) = \dot{x}$, so the field is tangent to the line.

A closed orbit must enclose at least one fixed point (the index of a closed orbit is $+1$). But every fixed point lies on one of these invariant lines, which extend to infinity; a line through a point in the interior of a closed orbit would have to cross the orbit, impossible since trajectories cannot cross invariant lines. Hence there are no closed orbits.

(c) The three invariant lines form a triangle with the fixed points as vertices: the saddle $(1, 0)$, the stable node $(-1, 0)$ and the unstable node $(2, 3)$. On $y = 0$: $\dot{x} = x^2 - 1$, so points with $|x| < 1$, and points with $x < -1$, flow to $(-1, 0)$, while points with $x > 1$ flow off to $+\infty$. Along the other two lines the flow leaves the unstable node, along $y = x + 1$ toward $(-1, 0)$ and along $y = 3(x - 1)$ toward the saddle $(1, 0)$ (and off to infinity on the far side of the node). The line $y = 3(x-1)$ is the separatrix: numerically, every trajectory starting above it is attracted to the stable node $(-1, 0)$, while every trajectory starting below it (to the right of the saddle) runs off to $x \to +\infty$. Inside the triangle, trajectories leave the unstable node and go to $(-1, 0)$. ■

7.2.15 *Consider $\dot{x} = x(2 - x - y)$, $\dot{y} = y(4x - x^2 - 3)$ (no closed orbits by Example 7.2.4). (a) Find the three fixed points and classify them. (b) Sketch the phase portrait.*

Solution. As in Example 7.2.4 we work in the closed positive quadrant $x, y \geq 0$.

(a) From $\dot{x} = 0$: $x = 0$ or $y = 2 - x$. From $\dot{y} = 0$: $y = 0$ or $x \in \{1, 3\}$. The solutions in the quadrant are $(0, 0)$, $(2, 0)$, $(1, 1)$. (There is a fourth fixed point $(3, -1)$, outside the quadrant.) The Jacobian is

$$J = \begin{pmatrix} 2 - 2x - y & -x \\ y(4 - 2x) & 4x - x^2 - 3 \end{pmatrix}.$$

- $(0, 0)$: $J = \operatorname{diag}(2, -3)$, *saddle* (unstable along the x-axis, stable along the y-axis).
- $(2, 0)$: $J = \begin{pmatrix} -2 & -2 \\ 0 & 1 \end{pmatrix}$, eigenvalues $-2, 1$: *saddle*, with stable direction along the x-axis and unstable eigenvector $(-2, 3)$, pointing into the quadrant.
- $(1, 1)$: $J = \begin{pmatrix} -1 & -1 \\ 2 & 0 \end{pmatrix}$, $\tau = -1$, $\Delta = 2$, $\tau^2 - 4\Delta = -7 < 0$: *stable spiral*.

(The fourth point $(3, -1)$ has $J = \begin{pmatrix} -3 & -3 \\ 2 & 0 \end{pmatrix}$: also a stable spiral, but it is not in the quadrant.)

(b) The axes are invariant. On the y-axis $\dot{y} = -3y$, flowing into the origin; on the x-axis $\dot{x} = x(2 - x)$, flowing from the origin to $(2, 0)$. The saddle at $(2, 0)$ sends its unstable manifold into the quadrant, and it ends at the spiral $(1, 1)$; since there are no closed orbits (Example 7.2.4) and trajectories are bounded, every trajectory in the open quadrant spirals into $(1, 1)$ (as a stable spiral, or nodelike, depending on the position), except the stable manifolds of the saddles. ■

7.2.16 *If R is not simply connected, the conclusion of Dulac's criterion can fail. Find a counterexample.*

Solution. Take $\dot{r} = r(1 - r^2)$, $\dot{\theta} = 1$, i.e. $\dot{x} = -y + x(1 - r^2)$, $\dot{y} = x + y(1 - r^2)$, and the annulus $R = \{0.8 < r < 1.2\}$ (not simply connected), with $g \equiv 1$. Then

$$\nabla \cdot \dot{\mathbf{x}} = (1 - r^2) - 2x^2 + (1 - r^2) - 2y^2 = 2 - 4r^2 < 2 - 4(0.64) < 0 \quad \text{on } R.$$

The divergence has a fixed sign on R, yet $r = 1$ is a closed orbit in R. The proof of Dulac's criterion breaks down because Green's theorem needs the whole interior of the closed orbit to lie in R; here the interior contains the hole. ■

7.2.17 *Assume the hypotheses of Dulac's criterion, except that R is topologically an annulus. Using Green's theorem show there is at most one closed orbit in R.*

Solution. Suppose $\nabla \cdot (g\dot{\mathbf{x}})$ has one sign (and is not identically zero) in R.

First, any closed orbit in R must encircle the hole: otherwise its interior lies in R and the usual Dulac argument gives a contradiction ($\iint \nabla \cdot (g\dot{\mathbf{x}})\, \mathrm{d}A \neq 0$ but the boundary integral $\oint g\dot{\mathbf{x}} \cdot \mathbf{n}\, \mathrm{d}l = 0$ since $\dot{\mathbf{x}}$ is tangent to the orbit).

Now suppose there were two closed orbits $C_1, C_2 \subset R$, both encircling the hole; being disjoint, one lies inside the other, say C_1 inside C_2. The region A between them lies in R (the only hole of R is inside C_1). Green's theorem (the divergence theorem) on A gives

$$\iint_A \nabla \cdot (g\dot{\mathbf{x}})\, \mathrm{d}A = \oint_{C_2} g\dot{\mathbf{x}} \cdot \mathbf{n}\, \mathrm{d}l - \oint_{C_1} g\dot{\mathbf{x}} \cdot \mathbf{n}\, \mathrm{d}l = 0 - 0 = 0,$$

because $\dot{\mathbf{x}} \cdot \mathbf{n} = 0$ on each orbit. But the integrand has a fixed sign and is not identically zero, so the integral is nonzero: contradiction. Hence there is at most one closed orbit. ■

7.2.18 *Consider* $\dot{x} = rx(1-x/2) - \dfrac{2x}{1+x}y$, $\dot{y} = -y + \dfrac{2x}{1+x}y$, $r>0$, $x,y \geq 0$. *Prove there are no closed orbits using Dulac with* $g = \dfrac{1+x}{x}y^{\alpha-1}$ *for a suitable* α.

Solution. Compute

$$g\dot{x} = y^{\alpha-1}\Big[r(1+x)\big(1-\tfrac{x}{2}\big) - 2y\Big], \qquad g\dot{y} = y^{\alpha}\Big[-\frac{1+x}{x} + 2\Big] = y^{\alpha}\,\frac{x-1}{x}.$$

Using $\frac{\mathrm{d}}{\mathrm{d}x}\big[(1+x)(1-\frac{x}{2})\big] = \frac{1}{2} - x$,

$$\nabla\cdot(g\dot{\mathbf{x}}) = y^{\alpha-1}\Big[r\big(\tfrac{1}{2}-x\big) + \alpha\,\frac{x-1}{x}\Big] = y^{\alpha-1}\Big[\tfrac{r}{2} + \alpha - rx - \frac{\alpha}{x}\Big].$$

Choose $\alpha = r$. Then the bracket is $r\big(\frac{3}{2} - x - \frac{1}{x}\big) \leq r\big(\frac{3}{2} - 2\big) = -\frac{r}{2} < 0$, because $x + 1/x \geq 2$ for $x > 0$. Hence $\nabla\cdot(g\dot{\mathbf{x}}) = -\,y^{r-1}r\big(x + \frac{1}{x} - \frac{3}{2}\big) < 0$ throughout the open quadrant $x, y > 0$, which is simply connected, with g smooth there. By Dulac's criterion there are no closed orbits in the open quadrant; none can lie on the axes either (they are invariant and carry no closed orbits: on $y = 0$, $\dot{x} = rx(1-x/2)$; on $x = 0$, $\dot{y} = -y$). ■

7.2.19 *(Love story in* Gone with the Wind*)* $\dot{R} = -R + A_S + kSe^{-S}$, $\dot{S} = -S + A_R + kRe^{-R}$, *with* $A_R, A_S, k > 0$. *(a) Interpret the terms. (b) Show the quadrant* $R, S \geq 0$ *is invariant; interpret. (c) Prove no periodic solutions by Dulac. (d) Plot the portrait for* $A_S = 1.2$, $A_R = 1$, $k = 15$ *and the trajectory from* $R(0) = S(0) = 0$.

Solution. *(a)* In $\dot{R} = -R + A_S + kSe^{-S}$: the term $-R$ is *forgetting*: Rhett's love decays exponentially in the absence of stimulus. The constant A_S is Rhett's response to Scarlett's inherent *appeal* (the parameter named after her), a steady source of attraction independent of how she feels about him. The third term, kSe^{-S}, is his *reaction to Scarlett's love for him.* The function Se^{-S} is increasing for $0 < S < 1$ (more affection from her draws him in), peaks at $S = 1$, and then decreases: when her love becomes too intense he is put off and loses interest. This is the "playing hard to get" (or fear of commitment) effect; Scarlett reacts to Rhett's love R in the same way, and A_R is Rhett's appeal to her. The symmetric roles are encoded in $\dot{S} = -S + A_R + kRe^{-R}$.

(b) On $R = 0$, $S \geq 0$: $\dot{R} = A_S + kSe^{-S} > 0$. On $S = 0$, $R \geq 0$: $\dot{S} = A_R + kRe^{-R} > 0$. So the vector field points strictly into the quadrant along both boundary axes, and the closed quadrant $R, S \geq 0$ is positively invariant: a trajectory that starts there stays there forever. Psychologically: if neither partner starts with negative feelings, neither ever develops hatred or resentment in this model (because each is attracted by the other's appeal); the relationship stays on the "love" side of indifference.

(c) Take $g \equiv 1$:

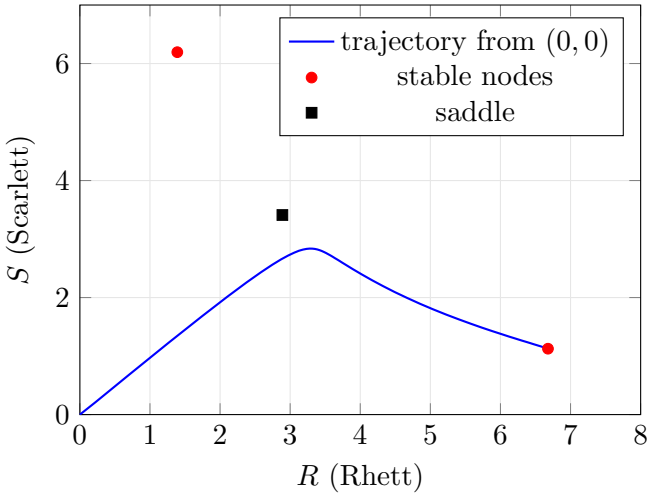
$$\nabla\cdot\dot{\mathbf{x}} = \frac{\partial\dot{R}}{\partial R} + \frac{\partial\dot{S}}{\partial S} = -1 - 1 = -2 < 0.$$

The plane is simply connected, so by Dulac's criterion there are no periodic solutions (anywhere in the plane).

(d) Numerically, for $A_S = 1.2$, $A_R = 1$, $k = 15$, in the first quadrant there are three fixed points:

$(R, S) = (1.390,\ 6.194)$	eigenvalues -0.52, -1.48	stable node
$(R, S) = (2.889,\ 3.411)$	eigenvalues $+0.37$, -2.37	saddle
$(R, S) = (6.678,\ 1.126)$	eigenvalues -0.74, -1.26	stable node

(there is a further saddle at $(-0.067, -0.078)$, outside the quadrant). The trajectory from $R(0) = S(0) = 0$ gives $(R, S) \approx (2.10, 2.02)$ at $t = 0.5$, $(2.97, 2.72)$ at $t = 1$, $(3.45, 2.80)$ at $t = 2$, $(5.41, 1.63)$ at $t = 6$, and settles at the stable node $(6.678, 1.126)$ (reached by $t \approx 15$). So the predicted "first stage": both start indifferent, are quickly drawn up together (each excited by the other's appeal, $R \approx S \approx 2$ within half a time unit), pass just below the saddle $(2.89, 3.41)$, and then Rhett's love keeps rising to $R \approx 6.7$ while Scarlett, put off by his intensity, backs off to $S \approx 1.1$: a lopsided steady state in which he loves her passionately and she is lukewarm. All trajectories in the quadrant go to one of the two stable nodes (the stable manifold of the saddle is the separatrix), since no periodic orbits exist.



■

3 Poincaré–Bendixson Theorem (7.3)

7.3.1 *Consider $\dot{x} = x - y - x(x^2 + 5y^2)$, $\dot{y} = x + y - y(x^2 + y^2)$. (a) Classify the origin. (b) Rewrite in polar coordinates. (c) Find the circle r_1 of maximum radius such that all trajectories point outward on it. (d) Find the circle r_2 of minimum radius such that all trajectories point inward on it. (e) Prove there is a limit cycle in $r_1 \le r \le r_2$.*

Solution. *(a)* Linearizing at the origin: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$, eigenvalues $1 \pm i$: an *unstable spiral*.

(b)

$$\begin{aligned} r\dot{r} &= x\dot{x} + y\dot{y} = x^2 - xy - x^2(x^2 + 5y^2) + xy + y^2 - y^2(x^2 + y^2) \\ &= r^2 - (x^4 + 6x^2y^2 + y^4). \end{aligned}$$

Since $\cos^4\theta + \sin^4\theta = 1 - 2\sin^2\theta\cos^2\theta$, we have $x^4 + 6x^2y^2 + y^4 = r^4(1 + 4\sin^2\theta\cos^2\theta) = r^4(1 + \sin^2 2\theta)$, so

$$\dot{r} = r - r^3\big(1 + \sin^2 2\theta\big).$$

For the angle: $x\dot{y} - y\dot{x} = r^2 + xy(x^2 + 5y^2 - x^2 - y^2) = r^2 + 4xy^3$, so

$$\dot{\theta} = 1 + 4r^2\cos\theta\sin^3\theta.$$

(c) We need $\dot r > 0$ for all θ: $1 > r^2(1+\sin^2 2\theta)$ for all θ, i.e. $r^2 < 1/\max(1+\sin^2 2\theta) = \frac12$. So $r_1 = 1/\sqrt{2} \approx 0.707$. (At $r = r_1$, $\dot r = \frac{r}{2}\cos^2 2\theta \geq 0$, vanishing only at isolated angles.)

(d) We need $\dot r < 0$ for all θ: $r^2(1+\sin^2 2\theta) > 1$ for all θ, i.e. $r^2 > 1/\min(1+\sin^2 2\theta) = 1$. So $r_2 = 1$. (At $r = 1$, $\dot r = -r\sin^2 2\theta \leq 0$.)

(e) By (c) and (d), trajectories cannot leave the annulus $A = \{r_1 \leq r \leq r_2\}$ (at the isolated points where $\dot r = 0$ on the boundary, argue as follows: $\dot r > 0$ strictly for $r < r_1$ and $\dot r < 0$ strictly for $r > r_2$, so a trajectory cannot cross out of A; alternatively enlarge A slightly to $r_1 - \delta \leq r \leq r_2 + \delta$, where the inequalities are strict). So A is a trapping region.

No fixed points except the origin. Let $a = x^2 + 5y^2$ and $b = x^2 + y^2$, so $a \geq b > 0$ and $a = b + 4y^2$. A fixed point satisfies $x - y = xa$ and $x + y = yb$, i.e. $y = x(1-a)$ and $x = -y(1-b)$. If $x = 0$ then $y = 0$. Otherwise substituting the first into the second gives $x = -x(1-a)(1-b)$, so

$$(1-a)(1-b) = -1.$$

Since $a \geq b$, this forces $1 - a < 0 < 1 - b$, i.e. $b < 1 < a$, and $(a-1)(1-b) = 1$. But

$$(a-1)(1-b) = -(1-b)^2 + 4y^2(1-b) \leq -(1-b)^2 + 4b(1-b) = (1-b)(5b-1) \leq 0.8 < 1$$

for $0 < b < 1$ (the maximum of $(1-b)(5b-1)$ is 0.8 at $b = 0.6$). This is a contradiction, so the origin is the only fixed point, and it is not in A.

By the Poincaré–Bendixson theorem, A contains a closed orbit (a limit cycle). ■

7.3.2 *Using numerical integration, compute the limit cycle of 7.3.1 and verify it lies in the trapping region.*

Solution. Integrating from $(0.9, 0)$ and from $(0.05, 0)$ (inside and outside the cycle) gives the same attracting closed orbit. Over the final cycles, $0.7228 \leq r \leq 0.8965$, which lies inside the trapping region $0.7071 \leq r \leq 1$, as it must. The period is $T \approx 10.476$ (longer than 2π because $\dot\theta = 1 + 4r^2\cos\theta\sin^3\theta$ drops to about 0.2 on parts of the orbit). The cycle is stable. ■

7.3.3 *Show that $\dot x = x - y - x^3$, $\dot y = x + y - y^3$ has a periodic solution.*

Solution. $r\dot r = x\dot x + y\dot y = r^2 - (x^4 + y^4)$, and $x^4 + y^4 = r^4(1 - \frac12\sin^2 2\theta) \in [\frac12 r^4, r^4]$. Hence

$$r - r^3 \leq \dot r \leq r - \tfrac12 r^3.$$

So $\dot r > 0$ for $r < 1$ and $\dot r < 0$ for $r > \sqrt{2}$. Also $x\dot y - y\dot x = r^2 + xy(x^2 - y^2)$, so $\dot\theta = 1 + \frac{r^2}{4}\sin 4\theta$.

No fixed points except the origin: at a fixed point with $r > 0$, $\dot r = 0$ forces $r^2 = x^4 + y^4 \leq r^4$ and $x^4 + y^4 \geq \frac12 r^4$, i.e. $1 \leq r^2 \leq 2$, while $\dot\theta = 0$ forces $\sin 4\theta = -4/r^2$, which needs $r^2 \geq 4$. Contradiction.

So the annulus $\frac12 \leq r \leq 2$ (say) is a trapping region (flow enters on both circles) without fixed points, and the Poincaré–Bendixson theorem gives a periodic solution in it. ■

7.3.4 *Consider $\dot x = x(1 - 4x^2 - y^2) - \frac12 y(1+x)$, $\dot y = y(1 - 4x^2 - y^2) + 2x(1+x)$. (a) Show the origin is unstable. (b) By considering $\dot V$ with $V = (1 - 4x^2 - y^2)^2$ show all trajectories approach the ellipse $4x^2 + y^2 = 1$.*

Solution. *(a)* Linearization at the origin: $\dot x \approx x - \frac12 y$, $\dot y \approx y + 2x$, matrix $\begin{pmatrix} 1 & -\frac12 \\ 2 & 1 \end{pmatrix}$, $\tau = 2$, $\Delta = 2$, eigenvalues $1 \pm i$: an unstable spiral.

(b) Let $q = 4x^2 + y^2$ and $E = 1 - q$, so $V = E^2$. Then

$$\begin{aligned}\dot{q} &= 8x\dot{x} + 2y\dot{y} = 8x^2E - 4xy(1+x) + 2y^2E + 4xy(1+x)\\ &= E\,(8x^2 + 2y^2) = 2q\,(1-q),\end{aligned}$$

since the cross terms cancel and $8x^2 + 2y^2 = 2q$. Hence $\dot{V} = -2E\dot{q} = -4q(1-q)^2 = -4q\,V \le 0$, with equality only on the ellipse $q = 1$ or at the origin.

The quantity q obeys the logistic equation $\dot{q} = 2q(1-q)$ *by itself*, so for every initial condition with $q(0) > 0$ (i.e. every point except the origin), $q(t) \to 1$, i.e. every trajectory approaches the ellipse $4x^2 + y^2 = 1$ as $t \to \infty$. (Explicitly, $q(t) = 1/\big(1 + (1/q_0 - 1)e^{-2t}\big)$.) ■

7.3.5 *Show that $\dot{x} = -x - y + x(x^2 + 2y^2)$, $\dot{y} = x - y + y(x^2 + 2y^2)$ has at least one periodic solution.*

Solution. $r\dot{r} = -r^2 + r^2(x^2 + 2y^2)$, so $\dot{r} = r\big(-1 + r^2(1 + \sin^2\theta)\big)$. The angular equation: $x\dot{y} - y\dot{x} = x^2 + y^2$ (the cubic terms cancel), so $\dot{\theta} = 1$. Since $1 \le 1 + \sin^2\theta \le 2$: $\dot{r} < 0$ for $r < 1/\sqrt{2}$ and $\dot{r} > 0$ for $r > 1$. Thus the annulus $\frac{1}{\sqrt{2}} \le r \le 1$ is a trapping region for the *time-reversed* flow ($t \to -t$), it contains no fixed point ($\dot{\theta} = 1 \neq 0$), so by Poincaré–Bendixson (applied to the reversed system) it contains a closed orbit. A closed orbit of the reversed flow is a closed orbit of the original system (an unstable one). Hence there is at least one periodic solution. ■

7.3.6 *Consider $\ddot{x} + F(x,\dot{x})\dot{x} + x = 0$ with $F < 0$ for $r \le a$ and $F > 0$ for $r \ge b$, $r^2 = x^2 + \dot{x}^2$. (a) Interpret physically. (b) Show there is at least one closed orbit with $a < r < b$.*

Solution. *(a)* F is a damping coefficient. $F < 0$ for small amplitude means negative damping: energy is pumped into small oscillations. $F > 0$ for large amplitude is ordinary dissipation. So small oscillations grow and large ones decay: this is the mechanism of the van der Pol oscillator (self-sustained oscillation).

(b) With $y = \dot{x}$: $\dot{x} = y$, $\dot{y} = -x - Fy$, so $r\dot{r} = xy + y(-x - Fy) = -Fy^2$, i.e. $\dot{r} = -Fr\sin^2\theta$. Thus $\dot{r} \ge 0$ for $r \le a$ and $\dot{r} \le 0$ for $r \ge b$, with equality only where $y = 0$. The only fixed point is the origin ($y = 0$ and then $x = 0$).

The annulus $a \le r \le b$ is trapping: a trajectory cannot cross from $r \ge a$ to $r < a$, since if it did, take the last time s with $r(s) = a$; on the subsequent interval $r < a$, so r^2 is nondecreasing there and cannot end below $r(s) = a$. Similarly one cannot cross from $r \le b$ to $r > b$. The annulus contains no fixed point, so by Poincaré–Bendixson it contains a closed orbit. ■

7.3.7 *Consider $\dot{x} = y + ax(1 - 2b - r^2)$, $\dot{y} = -x + ay(1 - r^2)$, $0 < a \le 1$, $0 \le b < \frac{1}{2}$. (a) Polar form. (b) Prove at least one limit cycle, and that all limit cycles have the same period $T(a,b)$. (c) Prove uniqueness for $b = 0$.*

Solution. *(a)* $r\dot{r} = xy + ax^2(1 - 2b - r^2) - xy + ay^2(1 - r^2) = ar^2\big[(1 - r^2) - 2b\cos^2\theta\big]$, so

$$\dot{r} = ar\big(1 - r^2 - 2b\cos^2\theta\big).$$

Also $x\dot{y} - y\dot{x} = -r^2 + 2abxy$, so

$$\dot{\theta} = -1 + ab\sin 2\theta.$$

(b) Since $ab < \frac{1}{2}$, $\dot{\theta} \le -1 + ab < -\frac{1}{2} < 0$: the angle decreases monotonically, there are no fixed points except the origin, and every closed orbit winds exactly once around the origin (clockwise). Radially, $\dot{r} > 0$ when $r^2 < 1 - 2b$ (as $1 - r^2 - 2b\cos^2\theta \ge 1 - r^2 - 2b > 0$; here $b < \frac{1}{2}$ makes $1 - 2b > 0$) and $\dot{r} < 0$ when $r > 1$. So the annulus $\sqrt{1 - 2b} \le r \le 1$ is a trapping region with no fixed points, and Poincaré–Bendixson gives at least one limit cycle.

Because $\dot\theta$ depends on θ only, the time to go once around is the same for every closed orbit:

$$T(a,b)=\int_0^{2\pi}\frac{\mathrm{d}\theta}{|\dot\theta|}=\int_0^{2\pi}\frac{\mathrm{d}\theta}{1-ab\sin 2\theta}=\frac{2\pi}{\sqrt{1-a^2b^2}},$$

using $\int_0^{2\pi}\mathrm{d}\phi/(1-k\sin\phi)=2\pi/\sqrt{1-k^2}$ (and the substitution $\phi=2\theta$, which covers two periods).

(c) For $b=0$, $\dot r=ar(1-r^2)$ is independent of θ, with the single positive root $r=1$, so $r=1$ is the only closed orbit (and it is stable). ■

7.3.8 *For $\dot r=r(1-r^2)+\mu r\cos\theta$, $\dot\theta=1$ (Example 7.3.1), plot for various $\mu>0$. Is there a critical μ_c at which the closed orbit ceases to exist? If not, prove an orbit exists for all $\mu>0$.*

Solution. There is *no* critical value: a closed orbit exists for every $\mu>0$. Since $\dot\theta=1$, use θ as the independent variable: $\mathrm{d}r/\mathrm{d}\theta=r(1-r^2+\mu\cos\theta)$, a Bernoulli equation. Put $u=r^{-2}$. Then

$$\frac{\mathrm{d}u}{\mathrm{d}\theta}=-2r^{-3}\frac{\mathrm{d}r}{\mathrm{d}\theta}=-2u\,(1+\mu\cos\theta)+2,$$

a *linear* equation. Its integrating factor is $e^{2\theta+2\mu\sin\theta}$, and the unique 2π-periodic solution is

$$u(\theta)=2\int_{-\infty}^{\theta}\exp\big[-2(\theta-s)-2\mu(\sin\theta-\sin s)\big]\,\mathrm{d}s>0.$$

(One checks directly that it is 2π-periodic and positive, being an integral of a positive function.) Hence $r(\theta)=u(\theta)^{-1/2}$ is a positive 2π-periodic function: a closed orbit exists for *every* μ, and it is unique, because any other solution of the linear equation differs from this one by $Ce^{-2\theta-2\mu\sin\theta}$ (so it is not periodic) and the Floquet multiplier is $e^{-4\pi}<1$, so the orbit is stable.

What does fail for $\mu\ge 1$ is the annulus $1-\mu\le r\le 1+\mu$ of Example 7.3.1 (its inner radius is no longer positive). The orbit is still there but it comes closer and closer to the origin as μ grows. Numerically: $r_{\min},r_{\max}=0.82,1.17$ at $\mu=0.4$; $0.52,1.40$ at $\mu=1$; $0.15,1.71$ at $\mu=2$; $6.7\times10^{-4},2.30$ at $\mu=5$; $4\times10^{-8},2.82$ at $\mu=10$. ■

7.3.9 *(Series approximation of the closed orbit) (a) For $\mu\ll1$ write $r(\theta)=1+\mu r_1(\theta)+O(\mu^2)$, substitute in the $\mathrm{d}r/\mathrm{d}\theta$ equation, neglect $O(\mu^2)$ terms, derive and solve an equation for r_1. (b) Find the max and min of r on your approximation and show they lie in $1-\mu\le r\le1+\mu$. (c) Compare with numerics; how does the max error depend on μ?*

Solution. *(a)* With $\mathrm{d}r/\mathrm{d}\theta=r(1-r^2+\mu\cos\theta)$ and $r=1+\mu r_1+O(\mu^2)$: the left side is $\mu r_1'$; on the right, $r^2=1+2\mu r_1+O(\mu^2)$, so $1-r^2+\mu\cos\theta=\mu(\cos\theta-2r_1)+O(\mu^2)$ and the right side is $\mu(\cos\theta-2r_1)+O(\mu^2)$. Equating the $O(\mu)$ terms,

$$r_1'+2r_1=\cos\theta.$$

The periodic solution is $r_1=A\cos\theta+B\sin\theta$ with $-A\sin\theta+B\cos\theta+2A\cos\theta+2B\sin\theta=\cos\theta$, i.e. $B+2A=1$, $2B-A=0$: $A=\frac25$, $B=\frac15$. Hence

$$r(\theta)\approx1+\frac{\mu}{5}\big(2\cos\theta+\sin\theta\big).$$

(b) $2\cos\theta+\sin\theta=\sqrt5\cos(\theta-\theta_0)$, so $r_{\max}\approx1+\mu/\sqrt5$ and $r_{\min}\approx1-\mu/\sqrt5$, with $1/\sqrt5\approx0.447<1$. So the approximate orbit lies in $1-\mu\le r\le1+\mu$, as expected.

(c) Comparing with the exact orbit from 7.3.8:

μ	0.4	0.2	0.1	0.05	0.025
max error $\max_\theta \lvert r_{\text{exact}} - r_{\text{approx}}\rvert$	0.0184	0.00429	0.00104	0.000254	0.0000630
error$/\mu^2$	0.115	0.107	0.1035	0.1017	0.1009

The maximum error is $\approx 0.10\,\mu^2$, i.e. it scales as $O(\mu^2)$, as the regular perturbation expansion predicts. ■

7.3.10 *Consider $\dot{\mathbf{x}} = A\mathbf{x} - r^2\mathbf{x}$, $r = \|\mathbf{x}\|$, A a real 2×2 matrix with complex eigenvalues $\alpha \pm i\omega$. Prove there is at least one limit cycle for $\alpha > 0$ and none for $\alpha \le 0$.*

Solution. Since A has non-real eigenvalues $\alpha \pm i\omega$ ($\omega \neq 0$), there is a real invertible P with $P^{-1}AP = J = \begin{pmatrix} \alpha & -\omega \\ \omega & \alpha \end{pmatrix}$. Put $\mathbf{x} = P\mathbf{z}$. Then

$$\dot{\mathbf{z}} = J\mathbf{z} - (\mathbf{z}^T M\mathbf{z})\,\mathbf{z}, \qquad M = P^TP \text{ (symmetric positive definite)},$$

because $r^2 = \|P\mathbf{z}\|^2 = \mathbf{z}^TM\mathbf{z}$. In polar coordinates (ρ,φ) for $\mathbf{z}$, $J\mathbf{z}$ contributes $\alpha\rho$ radially and $\omega\rho$ tangentially, while the cubic term is purely radial. So

$$\dot\rho = \rho\big(\alpha - \rho^2 m(\varphi)\big), \qquad \dot\varphi = \omega,$$

where $m(\varphi) = (\cos\varphi,\sin\varphi)M(\cos\varphi,\sin\varphi)^T \in [\lambda_{\min}(M),\lambda_{\max}(M)]$ with $0 < \lambda_{\min} \le \lambda_{\max}$.

Case $\alpha > 0$. $\dot\rho > 0$ for $\rho^2 < \alpha/\lambda_{\max}$ and $\dot\rho < 0$ for $\rho^2 > \alpha/\lambda_{\min}$. So the annulus $\alpha/\lambda_{\max} \le \rho^2 \le \alpha/\lambda_{\min}$ is trapping. Since $\dot\varphi = \omega \neq 0$, the only fixed point is the origin, not in the annulus. By Poincaré–Bendixson there is a limit cycle.

Case $\alpha \le 0$. For $\rho > 0$, $\dot\rho = \rho(\alpha - \rho^2 m) \le -\lambda_{\min}\rho^3 < 0$, so ρ strictly decreases along every trajectory; a closed orbit would require ρ to return to its initial value. Hence there are no closed orbits. (All trajectories spiral into the origin.) ■

7.3.11 *(Cycle graphs) (a) Plot the phase portrait of $\dot r = r(1-r^2)G$, $\dot\theta = G$, $G = r^2\sin^2\theta + (r^2\cos^2\theta - 1)^2$. (b) Sketch x vs. t for a trajectory starting away from the unit circle. What happens as $t\to\infty$?*

Solution. *(a)* In Cartesian terms $G = y^2 + (x^2-1)^2 \ge 0$, with $G = 0$ only at the two points $(\pm1, 0)$. Away from these points $G > 0$, so the system is just $\dot r = r(1-r^2)$, $\dot\theta = 1$ (the system of Section 7.1) with time rescaled by the positive factor G: the orbits are the same curves. Hence:

- the origin ($G(0,0) = 1$) is an unstable spiral, and trajectories inside the unit circle spiral counterclockwise outward toward the unit circle;
- trajectories outside spiral inward toward the unit circle;
- on $r = 1$: $G = \sin^2\theta(1+\sin^2\theta) \ge 0$, so $\dot\theta \ge 0$ with zeros exactly at $\theta = 0, \pi$. The unit circle therefore consists of the *two fixed points* $(1,0)$ and $(-1,0)$ connected by the *two semicircular trajectories* (upper and lower), both oriented counterclockwise. This is a cycle graph (heteroclinic cycle). Each of the fixed points is semistable along the circle ($\dot\theta \approx \theta^2$ near $\theta = 0$).

So there is no limit cycle: the unit circle is the ω-limit set of every trajectory except the origin, but it is not a closed orbit.

(b) For a trajectory starting off the unit circle, $r(t) \to 1$ and $\theta(t)$ increases, so $x = r\cos\theta$ oscillates with amplitude approaching 1; but the trajectory passes closer and closer to $(\pm1,0)$ where the speed is $O(G) \to 0$, and it dwells there for longer and longer. So $x(t)$ looks like an

oscillation between near $+1$ and near -1 with *plateaus* near $x = \pm 1$ whose durations grow without bound: the period tends to infinity (a "square wave" that slows down), and in the limit the solution creeps toward $x = \pm 1$ ever more slowly. As $t \to \infty$ it does not converge to any periodic function. ■

7.3.12 *(Heteroclinic cycle in rock–paper–scissors)* $\dot P = P[(aR-S)-(a-1)Q]$, $\dot R = R[(aS-P)-(a-1)Q]$, $\dot S = S[(aP-R)-(a-1)Q]$, $Q = PR+RS+PS$, $a>0$. $E_1 = P+R+S$, $E_2 = PRS$.

Solution. *(a)* Adding the three equations,

$$\dot E_1 = (aPR-PS+aRS-PR+aPS-RS)-(a-1)QE_1 = (a-1)Q-(a-1)QE_1 = (1-E_1)(a-1)Q.$$

So $\frac{\mathrm d}{\mathrm dt}(1-E_1) = -(a-1)Q\,(1-E_1)$, whence $1-E_1(t) = (1-E_1(0))\exp\bigl(-(a-1)\int_0^t Q\bigr)$: if $E_1(0)=1$ then $E_1 \equiv 1$. So the set $\{P+R+S=1\}$ is invariant: it is a *plane* in $\mathbb{R}^3$.

(b) Each equation has a factor (P, R or S), so the coordinate planes $P=0$, $R=0$, $S=0$ are invariant, and so is the set where all of $P,R,S \ge 0$. Hence $T = \{P,R,S\ge 0,\ P+R+S=1\}$ is invariant: it is a closed *equilateral triangle* (the 2-simplex) with vertices $(1,0,0),(0,1,0),(0,0,1)$.

(c) The vertices are fixed points (at $(1,0,0)$: $Q=0$ and each of the three right-hand sides vanishes). On the edge $S=0$, $P+R=1$ and $Q=PR$:

$$\dot P = PR\,[a-(a-1)P] = PR\,(P+aR) > 0, \qquad \dot R = -PR\,[1+(a-1)R] = -PR\,(P+aR) < 0$$

(using $P+R=1$). So along this edge the flow goes from the vertex $R=1$ to the vertex $P=1$. On the edge $R=0$: $\dot P = -PS\,(S+aP) < 0$, $\dot S = PS\,(S+aP) > 0$: from $P=1$ to $S=1$. On the edge $P=0$: $\dot R = RS(R+aS) > 0$, $\dot S < 0$: from $S=1$ to $R=1$. Hence ∂T consists of three fixed points connected by three trajectories, all oriented the same way ($R\to P\to S\to R$): a cycle graph (heteroclinic cycle).

(d) $\dot E_2 = E_2\bigl(\dot P/P + \dot R/R + \dot S/S\bigr) = E_2\bigl[(a-1)E_1 - 3(a-1)Q\bigr] = (a-1)E_2(E_1-3Q)$. On T, $E_1=1$, and

$$1-3Q = (P+R+S)^2-3(PR+RS+PS) = P^2+R^2+S^2-PR-RS-PS = \tfrac12\bigl[(P-R)^2+(R-S)^2+(S-P)^2\bigr],$$

so $\dot E_2 = \dfrac{(a-1)E_2}{2}\bigl[(P-R)^2+(R-S)^2+(S-P)^2\bigr]$.

(e) On ∂T at least one of P,R,S is 0, so $E_2=0$ and $\dot E_2 = 0$. At the interior fixed point $(P^*,R^*,S^*) = \frac13(1,1,1)$ (it is a fixed point: $aR-S = (a-1)/3 = (a-1)Q$ there) all three squares vanish, so $\dot E_2 = 0$. Elsewhere in the interior $E_2>0$ and the bracket is positive, so $\dot E_2$ has the sign of $a-1$.

(f) Let $a>1$. Then E_2 strictly increases along every interior trajectory other than the centre, and $E_2 \le \frac{1}{27}$ on T (AM–GM), with equality only at the centre. Take an interior trajectory; its ω-limit set is a nonempty compact invariant subset of T on which $\dot E_2 = 0$ (La Salle), so it lies in $\partial T \cup \{\text{centre}\}$. But $E_2 \ge E_2(0) > 0$ along the trajectory, so the limit set cannot touch ∂T (where $E_2=0$). Hence it is the centre: the interior fixed point attracts all trajectories in the interior of T.

(g) Let $a<1$. Now E_2 strictly decreases along interior trajectories (other than the centre), and by the same argument their ω-limit sets lie in $\partial T \cup \{\text{centre}\}$. The centre is excluded: a trajectory starting at a non-central point has $E_2(0) < \frac{1}{27}$ and E_2 decreases, so it cannot converge to the centre, where $E_2 = \frac{1}{27}$. So the limit set is a nonempty closed invariant connected subset of ∂T. It cannot be a single vertex: linearizing at the vertex $(1,0,0)$ inside T (coordinates R,S) gives $\dot R \approx -R$, $\dot S \approx aS$, a saddle whose stable manifold within T is the edge $S=0$ only, so no interior trajectory converges to it. Hence the limit set contains an edge point and therefore the whole boundary: the heteroclinic cycle attracts every interior trajectory (except the centre). ■

4 Liénard Systems (7.4)

Liénard's theorem. If f is even, g is odd with $g(x) > 0$ for $x > 0$, and $F(x) = \int_0^x f$ has a single positive zero $x = a$, is negative on $(0, a)$, is positive and nondecreasing for $x > a$ with $F(x) \to \infty$, then $\ddot{x} + f(x)\dot{x} + g(x) = 0$ has exactly one closed orbit in the phase plane, and it is stable.

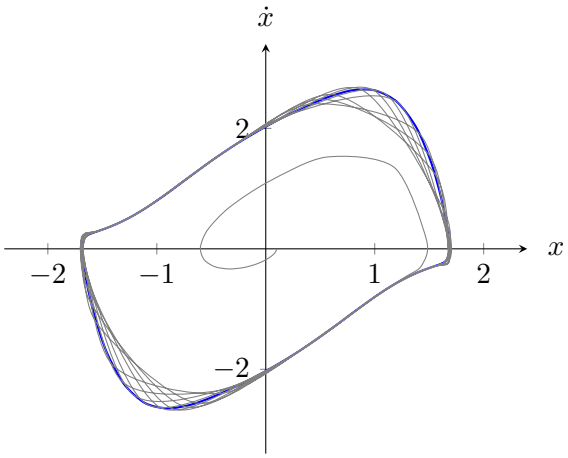
7.4.1 *Show $\ddot{x} + \mu(x^2 - 1)\dot{x} + \tanh x = 0$, $\mu > 0$, has exactly one periodic solution, and classify its stability.*

Solution. Here $f(x) = \mu(x^2 - 1)$ is even and $g(x) = \tanh x$ is odd with $g > 0$ for $x > 0$. Also $F(x) = \mu\big(\frac{1}{3}x^3 - x\big) = \frac{\mu}{3}x(x^2 - 3)$ has its only positive zero at $x = \sqrt{3}$, is negative on $(0, \sqrt{3})$, and is positive and increasing for $x > \sqrt{3}$ (since $F' = f > 0$ for $x > 1$), with $F \to \infty$. By Liénard's theorem there is exactly one periodic solution, and it is a *stable* limit cycle. ■

7.4.2 *Consider $\ddot{x} + \mu(x^4 - 1)\dot{x} + x = 0$. (a) Prove a unique stable limit cycle for $\mu > 0$. (b) Plot the portrait for $\mu = 1$. (c) If $\mu < 0$, is there still a limit cycle? Stable or unstable?*

Solution. *(a)* $f(x) = \mu(x^4 - 1)$ is even, $g(x) = x$ is odd and positive for $x > 0$. $F(x) = \mu\big(\frac{1}{5}x^5 - x\big)$ vanishes at $x = 0$ and $x = 5^{1/4} \approx 1.495$ (its only positive zero), is negative on $(0, 5^{1/4})$, and for $x > 5^{1/4} > 1$ it is positive and increasing since $F' = f > 0$ there, with $F \to \infty$. By Liénard's theorem there is a unique limit cycle, and it is stable.

(b) Numerically ($\mu = 1$, written as $\dot{x} = y$, $\dot{y} = -x - (x^4 - 1)y$), trajectories from $(0.1, 0)$, $(3, 0)$, $(0, 3)$ all converge to the same cycle with $x_{\max} \approx 1.684$, $y_{\max} = -y_{\min} \approx 2.649$ and period $T \approx 7.069$. The origin is an unstable spiral (linearization $\ddot{x} - \mu\dot{x} + x = 0$ has $\tau = 1, \Delta = 1$). Trajectories spiral out from the origin and in from far away to the single cycle; the cycle is somewhat distorted from a circle (it is squarer than the van der Pol cycle at the same μ).



(Blue: the limit cycle. Grey: the trajectory spiralling out from $(0.1, 0)$.)

(c) For $\mu < 0$ write $\mu = -|\mu|$ and let $s = -t$. Then $\mathrm{d}^2x/\mathrm{d}s^2 = \ddot{x}$ and $\mathrm{d}x/\mathrm{d}s = -\dot{x}$, so the equation becomes $x'' + |\mu|(x^4 - 1)x' + x = 0$, which is the $\mu > 0$ equation with $|\mu|$. It has a unique stable limit cycle in the variable s, i.e. the original system has a unique limit cycle which is stable under backward time, hence *unstable* in forward time. (The origin is then a stable spiral, and trajectories outside the cycle go off to infinity.) ■

5 Relaxation Oscillations (7.5)

7.5.1 *For the van der Pol oscillator with $\mu \gg 1$, show the positive branch of the cubic nullcline begins at $x_A = 2$ and ends at $x_B = 1$.*

Solution. In the Liénard plane, $\dot{x} = \mu\,(y - F(x))$, $\dot{y} = -x/\mu$ with $F(x) = \frac{1}{3}x^3 - x$. For $\mu \gg 1$, x is fast and y is slow; motion on the stable branches of the cubic $y = F(x)$ is slow, with $\dot{y} = -x/\mu$. $F'(x) = x^2 - 1$ so the local minimum is at $x = 1$, $F(1) = -\frac{2}{3}$, and the local maximum is at $x = -1$, $F(-1) = \frac{2}{3}$. The stable branches are $|x| > 1$.

On the right branch ($x > 1$) the slow motion has $\dot{y} < 0$, so the trajectory slides down the branch and reaches its lower end at the knee $B = (1, -\frac{2}{3})$: $x_B = 1$. There it leaves the branch and jumps horizontally (fast, y essentially constant) to the left until it meets the left branch where $F(x) = -\frac{2}{3}$:

$$\tfrac{1}{3}x^3 - x + \tfrac{2}{3} = 0 \iff (x-1)^2(x+2) = 0 \Rightarrow x = -2.$$

It lands at $C = (-2, -\frac{2}{3})$ and climbs the left branch ($\dot{y} = -x/\mu > 0$) to the knee $(-1, \frac{2}{3})$, from which it jumps horizontally right until $F(x) = \frac{2}{3}$:

$$\tfrac{1}{3}x^3 - x - \tfrac{2}{3} = 0 \iff (x+1)^2(x-2) = 0 \Rightarrow x = 2.$$

It lands at $A = (2, \frac{2}{3})$, the start of the slow descent along the right branch. Hence the positive branch is traversed from $x_A = 2$ down to $x_B = 1$. ■

7.5.2 *Redo the analysis of the van der Pol oscillator for $\mu \gg 1$ in the standard phase plane $\dot{x} = y$, $\dot{y} = -x - \mu(x^2 - 1)y$. What is the advantage of the Liénard plane?*

Solution. In the standard plane the x-nullcline is the x-axis $y = 0$ and the y-nullcline is the curve $y = -x/[\mu(x^2 - 1)]$, which is $O(1/\mu)$ close to the axis away from $x = \pm 1$ and has vertical asymptotes at $x = \pm 1$. Since $\dot{y} = -\mu(x^2 - 1)\,[y - y^*(x)]$ with $y^* = -x/[\mu(x^2 - 1)]$, the variable y is the fast one: for $|x| > 1$ it relaxes at rate $\mu(x^2 - 1)$ onto the curve $y \approx y^*(x) = O(1/\mu)$, and then x drifts slowly with $\dot{x} = y^*$ (toward $x = 1$ if $x > 1$). For $|x| < 1$ the coefficient changes sign and y is repelled from the curve: y grows exponentially and carries the trajectory away. During such a jump $\dot{x} = y$ and $\dot{y} \approx -\mu(x^2 - 1)y$ give $\mathrm{d}y/\mathrm{d}x = -\mu(x^2 - 1)$, i.e. $y + \mu F(x) = \text{const}$. So the jumps are *not* straight lines in this plane; they are the cubic curves $y = -\mu F(x) + c$ with y ranging over an $O(\mu)$ interval ($|y|$ up to about $\frac{2}{3}\mu$ and more), while the slow parts are squeezed within $O(1/\mu)$ of the x-axis. The cycle therefore has two very different length scales ($x = O(1)$, $y = O(\mu)$ on the jumps; $y = O(1/\mu)$ on the slow branches), and matching the two pieces is awkward.

The *advantage of the Liénard plane* (x, w), $w = y/\mu + F(x)$, is that the fast and slow directions are separated cleanly and all variables are $O(1)$: the fast motion is exactly horizontal (up to $O(1/\mu^2)$), the slow manifold is simply the cubic nullcline $w = F(x)$, and the limit cycle is the neat figure of two slow branches joined by two horizontal jumps. All the quantities (jump landing points $x = \pm 2$, knees at $x = \pm 1$, the slow-time integral) can be read off from the geometry of the cubic. ■

7.5.3 *Estimate the period of the limit cycle of $\ddot{x} + k(x^2 - 4)\dot{x} + x = 1$ for $k \gg 1$.*

Solution. Let $F(x) = k\big(\frac{1}{3}x^3 - 4x\big)$ so that the equation is $\frac{\mathrm{d}}{\mathrm{d}t}(\dot{x} + F(x)) = 1 - x$. With $w = \dot{x} + F(x)$, $y = w/k$ and $f_1(x) = \frac{1}{3}x^3 - 4x$:

$$\dot{x} = k\,(y - f_1(x)), \qquad \dot{y} = \frac{1 - x}{k}.$$

The cubic nullcline $y = f_1(x)$ has $f_1' = x^2 - 4$, so knees at $x = \pm 2$: $f_1(2) = -\frac{16}{3}$, $f_1(-2) = \frac{16}{3}$. The y-nullcline is $x = 1$, which meets the cubic on the middle (unstable) branch, so the system relaxation-oscillates.

Jumps. From the right knee $(2, -\frac{16}{3})$ the trajectory jumps left to the left branch where $f_1(x) = -\frac{16}{3}$: $x^3 - 12x + 16 = (x-2)^2(x+4) = 0$, so it lands at $x = -4$. From the left knee $(-2, \frac{16}{3})$ it jumps right to $x = 4$. The jumps take negligible time $O(1/k)$.

Slow branches. On a branch $y = f_1(x)$, so $\mathrm{d}y = (x^2-4)\mathrm{d}x$ and $\mathrm{d}t = \mathrm{d}y/\dot{y} = k\,\dfrac{x^2-4}{1-x}\,\mathrm{d}x$.

Right branch, from $x = 4$ down to $x = 2$ (there $\dot{y} = (1-x)/k < 0$, as needed):

$$T_R = k\int_4^2 \frac{x^2-4}{1-x}\,\mathrm{d}x = k\int_2^4 \Big(x+1-\frac{3}{x-1}\Big)\mathrm{d}x = k\Big[\tfrac{x^2}{2} + x - 3\ln(x-1)\Big]_2^4 = k\,(8 - 3\ln 3).$$

Left branch, from $x = -4$ up to $x = -2$ (there $\dot{y} > 0$):

$$T_L = k\int_{-4}^{-2} \frac{x^2-4}{1-x}\,\mathrm{d}x = k\Big[-\tfrac{x^2}{2} - x + 3\ln|x-1|\Big]_{-4}^{-2} = k\,\big(4 - 3\ln\tfrac{5}{3}\big).$$

Total period:

$$T \approx T_R + T_L = k\,(12 - 3\ln 3 - 3\ln 5 + 3\ln 3) = k\,(12 - 3\ln 5) \approx 7.17\,k.$$

■

7.5.4 *(Piecewise-linear nullclines) $\ddot{x} + \mu f(x)\dot{x} + x = 0$, $f = -1$ for $|x| < 1$, $f = 1$ for $|x| \geq 1$. (a) Show the system is equivalent to $\dot{x} = \mu(y - F(x))$, $\dot{y} = -x/\mu$ with F piecewise linear. (b) Graph nullclines. (c) Show relaxation oscillations for $\mu \gg 1$ and plot the cycle. (d) Estimate the period.*

Solution. *(a)* Let $F(x) = \int_0^x f$. For $|x| \leq 1$: $F = -x$. For $x \geq 1$: $F = -1 + (x-1) = x - 2$; for $x \leq -1$: $F = 1 + (x+1) = x + 2$ (oddness). Define $y = \dot{x}/\mu + F(x)$. Then $\dot{x} = \mu(y - F(x))$ and

$$\dot{y} = \frac{\ddot{x}}{\mu} + f(x)\dot{x} = \frac{-\mu f\dot{x} - x}{\mu} + f\dot{x} = -\frac{x}{\mu}.$$

(b) The x-nullcline is the graph $y = F(x)$: the slope-1 line $y = x + 2$ for $x \leq -1$, the slope-(-1) segment $y = -x$ for $|x| \leq 1$, and $y = x - 2$ for $x \geq 1$ (an "N" made of straight lines, with corners at $(-1, 1)$ and $(1, -1)$). The y-nullcline is the vertical line $x = 0$. They meet at the origin, on the middle branch.

(c) Linearizing at the origin, $\dot{x} = \mu(x + y)$, $\dot{y} = -x/\mu$: $\tau = \mu > 0$, $\Delta = 1$, so the fixed point is unstable. The fast motion is horizontal, $\dot{x} = \mu(y - F(x))$; it is stable on the outer branches ($F' = 1 > 0$) and unstable on the middle one ($F' = -1$). So for $\mu \gg 1$ a trajectory is attracted to an outer branch and slides along it until the knee, then jumps. Right branch $y = x - 2$: slides down ($\dot{y} = -x/\mu < 0$) to the knee $(1, -1)$, then jumps left at $y = -1$ to the left branch $y = x + 2$ at $x = -3$, i.e. to $(-3, -1)$. It slides up the left branch to $(-1, 1)$, then jumps right at $y = 1$ to the right branch at $x = 3$, arriving at $(3, 1)$. The limit cycle is therefore the closed "parallelogram" with corners $(3,1) \to (1,-1) \to (-3,-1) \to (-1,1) \to (3,1)$: slow along the slanted sides, fast along the horizontal sides.

(d) On the right branch $y = x - 2$, $\mathrm{d}y = \mathrm{d}x$ and $\dot{y} = -x/\mu$, so $\mathrm{d}t = -\mu\,\mathrm{d}x/x$. From $x = 3$ to $x = 1$:

$$T_{\text{right}} = \mu\int_1^3 \frac{\mathrm{d}x}{x} = \mu\ln 3.$$

By symmetry the left branch takes the same time. So

$$T \approx 2\mu \ln 3 \approx 2.20\,\mu.$$

■

7.5.5 *Consider $\ddot{x} + \mu(|x| - 1)\dot{x} + x = 0$. Find the approximate period of the limit cycle for $\mu \gg 1$.*
Solution. $f(x) = |x| - 1$, so $F(x) = \int_0^x f = \frac{1}{2}x|x| - x$: for $x \geq 0$, $F = \frac{1}{2}x^2 - x$, with minimum $F(1) = -\frac{1}{2}$; for $x \leq 0$, $F = -\frac{1}{2}x^2 - x$ with maximum $F(-1) = \frac{1}{2}$. In the Liénard plane ($\dot{x} = \mu(y - F)$, $\dot{y} = -x/\mu$) the outer branches ($|x| > 1$) are stable, the middle one is unstable, and the fixed point at the origin lies on the middle branch.

Jumps. From the right knee $(1, -\frac{1}{2})$ jump left to where $F(x) = -\frac{1}{2}$ on the left branch: $-\frac{1}{2}x^2 - x = -\frac{1}{2} \Rightarrow x^2 + 2x - 1 = 0 \Rightarrow x = -1 - \sqrt{2}$. By symmetry the other jump lands at $x = 1 + \sqrt{2}$.

Slow branch. On the right branch, $\mathrm{d}y = F'(x)\mathrm{d}x = (x - 1)\mathrm{d}x$ and $\dot{y} = -x/\mu$, so $\mathrm{d}t = -\mu\dfrac{x - 1}{x}\mathrm{d}x$. From $x = 1 + \sqrt{2}$ down to $x = 1$:

$$T_{\text{right}} = \mu \int_1^{1+\sqrt{2}} \frac{x - 1}{x}\,\mathrm{d}x = \mu\big[x - \ln x\big]_1^{1+\sqrt{2}} = \mu\big(\sqrt{2} - \ln(1 + \sqrt{2})\big).$$

The left branch takes the same time. Therefore

$$T \approx 2\mu\big(\sqrt{2} - \ln(1 + \sqrt{2})\big) \approx 1.066\,\mu.$$

■

7.5.6 *(Biased van der Pol) $\ddot{x} + \mu(x^2 - 1)\dot{x} + x = a$, $\mu > 0$. (a) Find and classify fixed points. (b) Plot the nullclines in the Liénard plane; show that if they intersect on the middle branch, the fixed point is unstable. (c) For $\mu \gg 1$ show there is a stable limit cycle iff $|a| < a_c$. (d) Sketch the portrait for a slightly greater than a_c; show the system is excitable.*
Solution. *(a)* With $\dot{x} = y$, $\dot{y} = a - x - \mu(x^2 - 1)y$ the only fixed point is $(x^*, y^*) = (a, 0)$. The Jacobian is $\begin{pmatrix} 0 & 1 \\ -1 & -\mu(a^2 - 1) \end{pmatrix}$: $\tau = -\mu(a^2 - 1)$, $\Delta = 1 > 0$. So the fixed point is *stable* if $|a| > 1$, *unstable* if $|a| < 1$ (a spiral if $\mu^2(a^2 - 1)^2 < 4$, a node otherwise), and the linearization is a centre at $|a| = 1$ (Hopf point).

(b) In the Liénard plane with $F(x) = \frac{1}{3}x^3 - x$: $\dot{x} = \mu(y - F(x))$, $\dot{y} = (a - x)/\mu$. The nullclines are the cubic $y = F(x)$ and the vertical line $x = a$; they meet at $(a, F(a))$. The middle branch of the cubic is $|x| < 1$; if the intersection lies there ($|a| < 1$) then by (a) $\tau = -\mu(a^2 - 1) > 0$ and the fixed point is unstable. (Geometrically: the middle branch is repelling for the fast flow, and the slow flow can't stabilize it.)

(c) For $\mu \gg 1$: motion is fast and horizontal except near the cubic; the slow flow on a stable branch ($|x| > 1$) has $\dot{y} = (a - x)/\mu$, which is negative for $x > a$ and positive for $x < a$. If $|a| < 1$, the nullcline $x = a$ meets the middle branch; on the right branch $x > 1 > a$ so $\dot{y} < 0$ and the trajectory slides down to the knee $(1, -\frac{2}{3})$, jumps left to $x = -2$, slides up the left branch (since $x < -1 < a$ there, $\dot{y} > 0$) to the knee $(-1, \frac{2}{3})$, jumps to $x = 2$, and the cycle repeats: a stable relaxation oscillation (the fixed point is on the middle branch, hence unstable, and the Poincaré–Bendixson theorem applies). If instead $a > 1$, the slow flow on the right branch (starting at $x = 2$ or anywhere with $x > a$) slides down only until $x = a$ where $\dot{y} = 0$: the fixed point $(a, F(a))$ sits on the stable right branch and attracts the trajectory before it reaches the knee, so there is no cycle

(the fixed point is globally attracting). The case $a < -1$ is symmetric. Hence the limit cycle exists iff $|a| < a_c = 1$ (to leading order for $\mu \gg 1$).

(d) For a slightly greater than 1, the fixed point $(a, F(a))$ is on the right branch just beyond the knee $B = (1, -\frac{2}{3})$, with $F(a) \approx -\frac{2}{3} + (a-1)^2$. It is a globally attracting stable node/spiral (all trajectories end there). Near the knee $F(x) \approx F(1) + (x-1)^2$. A small perturbation of the state at the fixed point decays back. But a perturbation that pushes the state to the left of the knee by more than about $a - 1$ (to $x < 2 - a$, below the cubic: $y = F(a) < F(x)$) makes $\dot{x} < 0$, and the trajectory takes the fast horizontal jump to the left branch, climbs it to the knee at $(-1, \frac{2}{3})$, jumps back to the right branch at $x = 2$, slides down to the knee and finally returns to the fixed point: a large excursion through phase space. So small disturbances die out, while a disturbance bigger than a small threshold triggers a full "spike": the system is *excitable*. ■

7.5.7 *(Cell cycle)* $\dot{u} = b(v-u)(\alpha + u^2) - u$, $\dot{v} = c - u$, *with* $b \gg 1$, $\alpha \ll 1$, $8\alpha b < 1$. *(a) Sketch the nullclines. (b) Show relaxation oscillations for* $c_1 < c < c_2$, *find* c_1, c_2 *approximately (assuming* $8\alpha b \ll 1$*). (c) Show the system is excitable if* c *is slightly less than* c_1.

Solution. *(a)* The v-nullcline is the vertical line $u = c$. The u-nullcline is the curve

$$v = N(u) = u + \frac{u}{b(\alpha + u^2)}.$$

Its derivative is $N'(u) = 1 + \dfrac{\alpha - u^2}{b(\alpha + u^2)^2}$. Setting $N' = 0$ gives, with $s = u^2$, $bs^2 + (2\alpha b - 1)s + \alpha(1 + \alpha b) = 0$, whose discriminant is $(2\alpha b - 1)^2 - 4b\alpha(1 + \alpha b) = 1 - 8\alpha b > 0$ (this is exactly the stated condition). The roots are

$$s_\pm = \frac{(1 - 2\alpha b) \pm \sqrt{1 - 8\alpha b}}{2b}.$$

For $8\alpha b \ll 1$, $\sqrt{1 - 8\alpha b} \approx 1 - 4\alpha b$, so $s_- \approx \alpha$ and $s_+ \approx 1/b - 3\alpha \approx 1/b$. Thus N has a local maximum at $u_1 \approx \sqrt{\alpha}$, with $N(u_1) \approx \sqrt{\alpha} + \dfrac{1}{2b\sqrt{\alpha}}$, and a local minimum at $u_2 \approx 1/\sqrt{b}$, with $N(u_2) \approx 2/\sqrt{b}$. The curve starts at the origin with the very steep slope $1 + 1/(\alpha b)$, rises to the maximum, falls to the minimum, then rises again, approaching the line $v \approx u$: an N-shaped (cubic-like) curve with a steep left branch.

(b) Since $\partial\dot{u}/\partial v = b(\alpha + u^2) > 0$, $\dot{u} > 0$ above the u-nullcline and $\dot{u} < 0$ below it; hence a branch where $N' > 0$ is stable for the fast u-flow, and one with $N' < 0$ is unstable. So the branches $u < u_1$ and $u > u_2$ are stable, and the middle branch $u_1 < u < u_2$ is unstable. Because $b \gg 1$, u is fast and v slow ($\dot{v} = c - u$: v increases for $u < c$ and decreases for $u > c$). The fixed point $(c, N(c))$ is on the middle (unstable) branch iff $u_1 < c < u_2$, and then the dynamics is a relaxation oscillation: slow rise of v up the left branch to the knee at u_1, a fast jump right to the right branch (landing at $u \approx N(u_1) \approx 1/(2b\sqrt{\alpha})$), a slow decrease of v down the right branch to the knee at u_2, a fast jump left to the left branch (landing at $u \approx 2\alpha\sqrt{b}$), and so on. Hence relaxation oscillations occur for $c_1 < c < c_2$ with

$$c_1 \approx \sqrt{\alpha}, \qquad c_2 \approx \frac{1}{\sqrt{b}}.$$

(Since $8\alpha b < 1$ gives $\sqrt{\alpha} < 1/\sqrt{8b} < 1/\sqrt{b}$, the interval is nonempty. Corrections are of relative order αb.)

(c) If c is slightly less than c_1, the vertical nullcline $u = c$ meets the left, stable branch just below the knee, so the fixed point is a globally attracting stable node. A small kick (in u or v)

is damped. But a kick that moves the state to the right of the middle branch, i.e. beyond the unstable branch which lies only a distance $\approx c_1 - c$ away, puts it above the u-nullcline, so $\dot u > 0$ and the trajectory jumps to the right branch, slides down it to the knee at u_2, jumps back to the left branch and returns to the fixed point: a full excursion (a "mitotic spike"). So there is a threshold: the system is excitable. ■

6 Weakly Nonlinear Oscillators (7.6)

For $\ddot x + x + \varepsilon h(x,\dot x) = 0$ with $x = r\cos\theta$, $\dot x = -r\sin\theta$, $\theta = t + \phi$, the averaged equations (7.6.53) are

$$r' = \langle h\sin\theta\rangle\,, \qquad r\phi' = \langle h\cos\theta\rangle\,, \qquad \langle\,\cdot\,\rangle = \frac{1}{2\pi}\int_0^{2\pi}\cdot\,\mathrm{d}\theta,$$

with $' = \mathrm{d}/\mathrm{d}T$, $T = \varepsilon t$, and the frequency is $\omega = 1 + \varepsilon\phi'$. I use the standard averages $\langle\cos^2\rangle = \frac12$, $\langle\cos^4\rangle = \frac38$, $\langle\cos^6\rangle = \frac{5}{16}$, $\langle\cos^2\sin^2\rangle = \frac18$, and that odd powers average to zero.

7.6.1 *Show that if (7.6.7) is expanded as a power series in ε, we recover (7.6.17).*

Solution. (7.6.7) is $x(t,\varepsilon) = (1-\varepsilon^2)^{-1/2}e^{-\varepsilon t}\sin\big[(1-\varepsilon^2)^{1/2}t\big]$. Expand each factor: $(1-\varepsilon^2)^{-1/2} = 1 + \frac12\varepsilon^2 + \cdots$, $e^{-\varepsilon t} = 1 - \varepsilon t + \frac12\varepsilon^2t^2 - \cdots$, and $\sin[(1 - \frac12\varepsilon^2 + \cdots)t] = \sin t - \frac12\varepsilon^2 t\cos t + \cdots$. Multiplying, to $O(\varepsilon)$:

$$x(t,\varepsilon) = \sin t - \varepsilon t\sin t + O(\varepsilon^2),$$

which is (7.6.17). (To the next order: $+\varepsilon^2\big[\frac12\sin t + \frac12 t^2\sin t - \frac12 t\cos t\big]$.) ■

7.6.2 *(Calibrating regular perturbation theory) $\ddot x + x + \varepsilon x = 0$, $x(0) = 1$, $\dot x(0) = 0$. (a) Exact solution. (b) Find x_0, x_1, x_2 in $x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + O(\varepsilon^3)$. (c) Secular terms?*

Solution. *(a)* $\ddot x + (1+\varepsilon)x = 0$, so $x(t) = \cos\big(\sqrt{1+\varepsilon}\,t\big)$.

(b) Substituting the series and collecting powers:

$$O(1):\ \ddot x_0 + x_0 = 0; \qquad O(\varepsilon):\ \ddot x_1 + x_1 = -x_0; \qquad O(\varepsilon^2):\ \ddot x_2 + x_2 = -x_1,$$

with $x_0(0) = 1$, $\dot x_0(0) = 0$ and $x_k(0) = \dot x_k(0) = 0$ for $k \ge 1$. Then $x_0 = \cos t$. For x_1: $\ddot x_1 + x_1 = -\cos t$ is resonantly forced; the solution with zero initial data is $x_1 = -\frac12 t\sin t$ (check: $\ddot x_1 = -\cos t + \frac12 t\sin t$). Then $\ddot x_2 + x_2 = \frac12 t\sin t$, whose solution with zero initial data is

$$x_2 = \tfrac18 t\sin t - \tfrac18 t^2\cos t.$$

(To verify: a particular solution is $\mathrm{Im}\big[(\frac{t}{8} - \frac{i}{8}t^2)e^{it}\big]$, and it already satisfies $x_2(0) = \dot x_2(0) = 0$.) Thus

$$x = \cos t - \tfrac12\varepsilon\, t\sin t + \varepsilon^2\big(\tfrac18 t\sin t - \tfrac18 t^2\cos t\big) + O(\varepsilon^3).$$

This agrees with the expansion of the exact solution: $\sqrt{1+\varepsilon} = 1 + \frac{\varepsilon}{2} - \frac{\varepsilon^2}{8} + \cdots$ and $\cos(t+\delta) = \cos t - \delta\sin t - \frac12\delta^2\cos t$ with $\delta = (\frac{\varepsilon}{2} - \frac{\varepsilon^2}{8})t$.

(c) Yes: $t\sin t$ and $t^2\cos t$ are secular terms. They were to be expected: the perturbation changes the frequency (to $\sqrt{1+\varepsilon} \approx 1 + \varepsilon/2$), and a frequency shift appears in a fixed-t expansion in ε as $\cos((1+\varepsilon/2)t) \approx \cos t - \frac{\varepsilon}{2}t\sin t$. They are artefacts: the true solution is bounded, and the series is only good for $\varepsilon t \ll 1$. ■

7.6.3 *(More calibration) $\ddot x + x = \varepsilon$, $x(0) = 1$, $\dot x(0) = 0$. (a) Solve exactly. (b) Find x_0, x_1, x_2. (c) Explain why there are or are not secular terms.*

Solution. *(a)* $x = \varepsilon + (1-\varepsilon)\cos t$.

(b) $O(1)$: $\ddot{x}_0 + x_0 = 0 \Rightarrow x_0 = \cos t$. $O(\varepsilon)$: $\ddot{x}_1 + x_1 = 1$, $x_1(0) = \dot{x}_1(0) = 0 \Rightarrow x_1 = 1 - \cos t$. $O(\varepsilon^2)$: $\ddot{x}_2 + x_2 = 0$ with zero initial data $\Rightarrow x_2 = 0$. So $x = \cos t + \varepsilon(1-\cos t) + 0\cdot\varepsilon^2 + \cdots$, which reproduces the exact solution.

(c) There are no secular terms: the forcing at $O(\varepsilon)$ is a constant, which is not resonant with the natural frequency 1 of $\ddot{x} + x = 0$, and the perturbation does not alter the frequency of oscillation (it only shifts the equilibrium to $x = \varepsilon$). So the true solution is exactly a polynomial in ε with bounded coefficients. ■

7.6.4 $h = x$.

Solution. $h = r\cos\theta$. $r' = \langle r\cos\theta\sin\theta\rangle = 0$; $r\phi' = \langle r\cos^2\theta\rangle = \frac{r}{2}$, so $\phi' = \frac{1}{2}$. Hence $r \equiv a$ and $\phi = \frac{1}{2}T$ (with $\phi(0) = 0$ from $x(0) = a$, $\dot{x}(0) = 0$). The frequency is $\omega = 1 + \frac{1}{2}\varepsilon + O(\varepsilon^2)$ and

$$x(t,\varepsilon) \approx a\cos\big[(1+\tfrac{\varepsilon}{2})t\big].$$

The equation is the linear oscillator $\ddot{x} + (1+\varepsilon)x = 0$ (exact frequency $\sqrt{1+\varepsilon} \approx 1 + \frac{\varepsilon}{2}$), a centre: all orbits are periodic, no limit cycles. ■

7.6.5 $h = x\dot{x}^2$.

Solution. $h = r^3\cos\theta\sin^2\theta$. $r' = r^3\langle\cos\theta\sin^3\theta\rangle = 0$. $r\phi' = r^3\langle\cos^2\theta\sin^2\theta\rangle = \frac{1}{8}r^3$, so $\phi' = \frac{1}{8}r^2$. Hence $r \equiv a$, $\phi = \frac{1}{8}a^2T$, and the frequency is

$$\omega = 1 + \tfrac{1}{8}\varepsilon a^2 + O(\varepsilon^2), \qquad x(t,\varepsilon) \approx a\cos\big[(1+\tfrac{1}{8}\varepsilon a^2)t\big].$$

There are no limit cycles; there is a continuous family of periodic orbits whose frequency increases with amplitude. ■

7.6.6 $h = x\dot{x}$.

Solution. $h = -r^2\cos\theta\sin\theta$. $r' = -r^2\langle\cos\theta\sin^2\theta\rangle = 0$ and $r\phi' = -r^2\langle\cos^2\theta\sin\theta\rangle = 0$. So $r \equiv a$, $\phi \equiv 0$: to this order the amplitude and frequency are unchanged, $x \approx a\cos t$, $\omega = 1 + O(\varepsilon^2)$. There are no limit cycles. Indeed the equation $\ddot{x} + x + \varepsilon x\dot{x} = 0$ is reversible (invariant under $t \to -t$, $x \to -x$), so the origin is a nonlinear centre and all nearby orbits are closed. ■

7.6.7 $h = (x^4 - 1)\dot{x}$.

Solution. $h = -(r^4\cos^4\theta - 1)\,r\sin\theta$.

$$r' = -r\langle(r^4\cos^4\theta - 1)\sin^2\theta\rangle = -r\Big[r^4\big(\langle\cos^4\rangle - \langle\cos^6\rangle\big) - \tfrac{1}{2}\Big] = -r\Big[\frac{r^4}{16} - \frac{1}{2}\Big] = \frac{r}{2}\Big(1 - \frac{r^4}{8}\Big),$$

using $\langle\cos^4\theta\sin^2\theta\rangle = \frac{3}{8} - \frac{5}{16} = \frac{1}{16}$. Also $r\phi' = -r\langle(r^4\cos^4\theta - 1)\sin\theta\cos\theta\rangle = 0$, so $\phi' = 0$.

Fixed points of r': $r = 0$ (unstable, $r' \approx r/2$) and $r^4 = 8$, i.e. $r^* = 2^{3/4} \approx 1.68$, which is stable ($r' > 0$ below, < 0 above). So there is a stable limit cycle of amplitude $2^{3/4}$ and frequency $\omega = 1 + O(\varepsilon^2)$ (the cycle is nearly circular with radius ≈ 1.68; compare numerics for $\mu = 1$ in 7.4.2, where the amplitude is 1.68).

Solving: $u = r^{-4}$ gives $u' = -4r^{-5}r' = -2u + \frac{1}{4}$, so $u = \frac{1}{8} + (a^{-4} - \frac{1}{8})e^{-2T}$ and

$$x(t,\varepsilon) \approx \Big[\tfrac{1}{8} + (a^{-4} - \tfrac{1}{8})e^{-2\varepsilon t}\Big]^{-1/4}\cos t.$$

■

7.6.8 $h = (|x| - 1)\dot{x}$.

Solution. $h = -(r|\cos\theta| - 1)\, r\sin\theta$, so

$$r' = -r\Big[r\,\big\langle |\cos\theta|\sin^2\theta\big\rangle - \big\langle \sin^2\theta\big\rangle\Big].$$

$\big\langle |\cos\theta|\sin^2\theta\big\rangle = \frac{4}{2\pi}\int_0^{\pi/2}\cos\theta\sin^2\theta\,\mathrm{d}\theta = \frac{2}{\pi}\cdot\frac{1}{3} = \frac{2}{3\pi}$. Hence

$$r' = -r\Big[\frac{2r}{3\pi} - \frac{1}{2}\Big] = \frac{r}{2}\Big(1 - \frac{4r}{3\pi}\Big).$$

Also $r\phi' = -r\,\big\langle (r|\cos\theta| - 1)\sin\theta\cos\theta\big\rangle = 0$ (the integrand is odd under $\theta \to -\theta$), so $\phi' = 0$. The origin is unstable, and $r^* = 3\pi/4 \approx 2.36$ is stable: a stable limit cycle of amplitude $3\pi/4$ and frequency $\omega = 1 + O(\varepsilon^2)$. The equation for r is logistic, $r' = \frac{1}{2}r(1 - r/r^*)$, so

$$r(T) = \frac{r^*}{1 + (r^*/a - 1)e^{-T/2}}, \qquad x(t,\varepsilon) \approx r(\varepsilon t)\cos t.$$

■

7.6.9 $h = (x^2 - 1)\dot{x}^3$.

Solution. $h = -(r^2\cos^2\theta - 1)\, r^3\sin^3\theta$.

$$r' = -r^3\,\big\langle (r^2\cos^2\theta - 1)\sin^4\theta\big\rangle = -r^3\Big[r^2\big(\big\langle\sin^4\big\rangle - \big\langle\sin^6\big\rangle\big) - \big\langle\sin^4\big\rangle\Big] = -r^3\Big[\frac{r^2}{16} - \frac{3}{8}\Big] = \frac{r^3}{16}\big(6 - r^2\big).$$

And $r\phi' = -r^3\,\big\langle (r^2\cos^2\theta - 1)\sin^3\theta\cos\theta\big\rangle = 0$, so $\phi' = 0$. The origin is unstable ($r' \approx \frac{3}{8}r^3$), and $r^* = \sqrt{6} \approx 2.45$ is stable. So there is a stable limit cycle of amplitude $\sqrt{6}$ and frequency $\omega = 1 + O(\varepsilon^2)$.

The amplitude equation is separable. With $w = r^2$, $\mathrm{d}w/\mathrm{d}T = \frac{1}{8}w^2(6 - w)$, and partial fractions $\dfrac{1}{w^2(6-w)} = \dfrac{1}{6w^2} + \dfrac{1}{36w} + \dfrac{1}{36(6-w)}$ give the implicit solution

$$-\frac{1}{6r^2} + \frac{1}{36}\ln\frac{r^2}{6 - r^2} = \frac{T}{8} + C, \qquad C = -\frac{1}{6a^2} + \frac{1}{36}\ln\frac{a^2}{6 - a^2},$$

which cannot be solved for r in elementary closed form. With $\phi \equiv 0$, $x \approx r(\varepsilon t)\cos t$. ■

7.6.10 *Derive* $\sin\theta\cos^2\theta = \frac{1}{4}[\sin\theta + \sin 3\theta]$ *with complex exponentials.*

Solution. $\cos^2\theta = \frac{1}{4}(e^{2i\theta} + 2 + e^{-2i\theta})$ and $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$, so

$$\begin{aligned}
\sin\theta\cos^2\theta &= \frac{1}{8i}\big(e^{i\theta} - e^{-i\theta}\big)\big(e^{2i\theta} + 2 + e^{-2i\theta}\big) \\
&= \frac{1}{8i}\big(e^{3i\theta} + 2e^{i\theta} + e^{-i\theta} - e^{i\theta} - 2e^{-i\theta} - e^{-3i\theta}\big) \\
&= \frac{1}{8i}\big[(e^{3i\theta} - e^{-3i\theta}) + (e^{i\theta} - e^{-i\theta})\big] = \frac{1}{8i}\big[2i\sin 3\theta + 2i\sin\theta\big] = \tfrac{1}{4}(\sin\theta + \sin 3\theta).
\end{aligned}$$

■

7.6.11 *(Higher harmonics) Return to Example 7.6.2 (van der Pol) and solve for* x_1*, assuming initial conditions* $x(0) = 2$, $\dot{x}(0) = 0$.

Solution. Two-timing with $\tau = t$, $T = \varepsilon t$, $x = x_0(\tau, T) + \varepsilon x_1(\tau, T) + \cdots$. The $O(1)$ equation gives $x_0 = r(T)\cos(\tau + \phi(T))$. At $O(\varepsilon)$, with $\theta = \tau + \phi$,

$$\frac{\partial^2 x_1}{\partial \tau^2} + x_1 = -2\frac{\partial^2 x_0}{\partial \tau \partial T} - (x_0^2 - 1)\frac{\partial x_0}{\partial \tau} = 2(r' \sin\theta + r\phi' \cos\theta) + (r^2\cos^2\theta - 1)\, r \sin\theta.$$

Using $\sin\theta\cos^2\theta = \frac{1}{4}(\sin\theta + \sin 3\theta)$ (7.6.10), the right side is

$$\big(2r' + \tfrac{1}{4}r^3 - r\big)\sin\theta + 2r\phi'\cos\theta + \tfrac{1}{4}r^3 \sin 3\theta.$$

Removing secular terms gives $r' = \frac{r}{2}(1 - \frac{r^2}{4})$, $\phi' = 0$ and leaves

$$\frac{\partial^2 x_1}{\partial \tau^2} + x_1 = \tfrac{1}{4}r^3 \sin 3\theta \implies x_1 = -\tfrac{1}{32}r^3 \sin 3\theta + A(T)\cos\theta + B(T)\sin\theta.$$

Initial conditions. At leading order $x_0(0,0) = r(0)\cos\phi(0) = 2$ and $\partial_\tau x_0(0,0) = -r(0)\sin\phi(0) = 0$, so $\phi(0) = 0$ and $r(0) = 2$: the initial condition lies on the limit cycle, so $r \equiv 2$ and $\phi \equiv 0$ for all T. Then $x_0 = 2\cos\tau$ and the third-harmonic coefficient is $-\frac{1}{32}\cdot 8 = -\frac{1}{4}$. At $O(\varepsilon)$ the conditions are $x_1(0,0) = 0$ and $\partial_\tau x_1(0,0) + \partial_T x_0(0,0) = 0$ with $\partial_T x_0 = 0$; these give $A(0) = 0$ and $-\frac{3}{4} + B(0) = 0$, i.e. $B(0) = \frac{3}{4}$. (The slow dependence of A, B on T is fixed at the next order; it does not affect the point being made here.) Hence

$$x_1(\tau) = \tfrac{3}{4}\sin\tau - \tfrac{1}{4}\sin 3\tau = \sin^3\tau,$$

$$x(t,\varepsilon) \approx 2\cos t + \varepsilon\Big(\tfrac{3}{4}\sin t - \tfrac{1}{4}\sin 3t\Big) + O(\varepsilon^2).$$

The third harmonic $\sin 3t$, absent in the linear oscillator, is generated by the nonlinearity with amplitude $\varepsilon/4$. ■

7.6.12 *(Fourier coefficients) Show* $\langle \cos k\theta \sin m\theta\rangle = 0$ *for all* k, m; $\langle \cos k\theta \cos m\theta\rangle = \langle \sin k\theta \sin m\theta\rangle = 0$ *for* $k \neq m$; $\langle \cos^2 k\theta\rangle = \langle \sin^2 k\theta\rangle = \frac{1}{2}$ *for* $k \neq 0$*; and derive (7.6.51).*

Solution. Take $k, m \geq 0$ integers (as in a Fourier series). Recall $\langle \cos n\theta\rangle = \langle \sin n\theta\rangle = 0$ for every integer $n \neq 0$ (the integral over a full period of a nonconstant sinusoid vanishes).

(a) By the product-to-sum formulas:

$$\begin{aligned}
\langle \cos k\theta \sin m\theta\rangle &= \tfrac{1}{2}\langle \sin(m+k)\theta + \sin(m-k)\theta\rangle = 0 \quad \text{(the average of a sine is 0, also if } m = k),\\
\langle \cos k\theta \cos m\theta\rangle &= \tfrac{1}{2}\langle \cos(k+m)\theta + \cos(k-m)\theta\rangle = 0 \quad (k \neq m,\ k+m \neq 0),\\
\langle \sin k\theta \sin m\theta\rangle &= \tfrac{1}{2}\langle \cos(k-m)\theta - \cos(k+m)\theta\rangle = 0 \quad (k \neq m,\ k+m \neq 0),
\end{aligned}$$

and for $k = m \neq 0$: $\langle \cos^2 k\theta\rangle = \frac{1}{2}\big(1 + \langle \cos 2k\theta\rangle\big) = \frac{1}{2}$, $\langle \sin^2 k\theta\rangle = \frac{1}{2}\big(1 - \langle \cos 2k\theta\rangle\big) = \frac{1}{2}$.

(b) Write $h(\theta) = \sum_{k\geq 0} a_k \cos k\theta + \sum_{k\geq 1} b_k \sin k\theta$ (7.6.50), multiply by $\cos m\theta$ ($m \geq 1$) and average term by term. By (a), every term vanishes except $a_m\langle \cos^2 m\theta\rangle = \frac{1}{2}a_m$. So $\langle h\cos m\theta\rangle = \frac{1}{2}a_m$, i.e.

$$a_m = 2\langle h\cos m\theta\rangle = \frac{1}{\pi}\int_0^{2\pi} h(\theta)\cos m\theta\, \mathrm{d}\theta.$$

(c) Multiplying by $\sin m\theta$ ($m \geq 1$) similarly gives $\langle h\sin m\theta\rangle = \frac{1}{2}b_m$, i.e. $b_m = \frac{1}{\pi}\int_0^{2\pi} h\sin m\theta\, \mathrm{d}\theta$. Averaging h itself (multiplying by $\cos 0\theta = 1$): $\langle h\rangle = a_0$, i.e. $a_0 = \frac{1}{2\pi}\int_0^{2\pi} h\, \mathrm{d}\theta$. These are the formulas (7.6.51). ■

7.6.13 *(Exact period of a conservative oscillator)* $\ddot{x}+x+\varepsilon x^3=0$, $x(0)=a$, $\dot{x}(0)=0$. *(a) Express the period* $T(\varepsilon)$ *as an integral. (b) Expand in* ε *to find* c_0, c_1, c_2 *in* $T=c_0+c_1\varepsilon+c_2\varepsilon^2+O(\varepsilon^3)$ *and check* c_0, c_1 *against (7.6.57).*

Solution. *(a)* Energy conservation: $\frac{1}{2}\dot{x}^2+\frac{1}{2}x^2+\frac{1}{4}\varepsilon x^4=\frac{1}{2}a^2+\frac{1}{4}\varepsilon a^4$, so

$$\dot{x}^2=(a^2-x^2)\Big[1+\tfrac{\varepsilon}{2}(a^2+x^2)\Big].$$

The period is four times the time to go from $x=a$ to $x=0$:

$$T(\varepsilon)=4\int_0^a \frac{\mathrm{d}x}{\sqrt{(a^2-x^2)\,[1+\frac{\varepsilon}{2}(a^2+x^2)]}}=4\int_0^{\pi/2}\frac{\mathrm{d}\psi}{\sqrt{1+\frac{\varepsilon a^2}{2}(1+\sin^2\psi)}},$$

with $x=a\sin\psi$.

(b) Let $z=\frac{\varepsilon a^2}{2}(1+\sin^2\psi)$ and $(1+z)^{-1/2}=1-\frac{z}{2}+\frac{3}{8}z^2-\cdots$. With $\int_0^{\pi/2}\mathrm{d}\psi=\frac{\pi}{2}$, $\int\sin^2=\frac{\pi}{4}$, $\int\sin^4=\frac{3\pi}{16}$: $\int(1+\sin^2)=\frac{3\pi}{4}$ and $\int(1+\sin^2)^2=\frac{\pi}{2}+2\cdot\frac{\pi}{4}+\frac{3\pi}{16}=\frac{19\pi}{16}$. Hence

$$T=4\Big[\frac{\pi}{2}-\frac{\varepsilon a^2}{4}\cdot\frac{3\pi}{4}+\frac{3}{8}\cdot\frac{\varepsilon^2a^4}{4}\cdot\frac{19\pi}{16}\Big]+O(\varepsilon^3)=2\pi-\frac{3\pi}{4}a^2\,\varepsilon+\frac{57\pi}{128}a^4\,\varepsilon^2+O(\varepsilon^3).$$

So $c_0=2\pi$, $c_1=-\frac{3\pi}{4}a^2$, $c_2=\frac{57\pi}{128}a^4$. Check: (7.6.57) says $\omega=1+\frac{3}{8}\varepsilon a^2$, so $T=2\pi/\omega\approx 2\pi\big(1-\frac{3}{8}\varepsilon a^2\big)=2\pi-\frac{3\pi}{4}a^2\varepsilon$, which agrees with c_0 and c_1. ■

7.6.14 *(Computer test of two-timing)* $\ddot{x}+\varepsilon\dot{x}^3+x=0$. *(a) Averaged equations. (b) Solve for* $x(0)=a$, $\dot{x}(0)=0$. *(c) Compare numerically for* $a=1$, $\varepsilon=2$, $0\le t\le 50$.

Solution. *(a)* $h=\dot{x}^3=-r^3\sin^3\theta$. $r'=-r^3\langle\sin^4\theta\rangle=-\frac{3}{8}r^3$ and $r\phi'=-r^3\langle\sin^3\theta\cos\theta\rangle=0$.

(b) $\phi\equiv 0$. With $u=r^{-2}$: $u'=-2r^{-3}r'=\frac{3}{4}$, so $u=a^{-2}+\frac{3}{4}T$ and

$$r(T)=\frac{a}{\sqrt{1+\frac{3}{4}a^2T}},\qquad x(t,\varepsilon)\approx\frac{a\cos t}{\sqrt{1+\frac{3}{4}\varepsilon a^2 t}}.$$

(c) For $a=1$, $\varepsilon=2$ the true solution (numerical integration) and the formula $x\approx\cos t/\sqrt{1+\frac{3}{2}t}$ agree well: the peak amplitudes agree to about three digits (e.g. at $t=3.49$: 0.4013 vs. 0.4003; $t=9.82$: 0.2523 vs. 0.2521; $t=22.4$: 0.1700 vs. 0.1700; $t=35.0$: 0.1368 vs. 0.1368). The largest pointwise discrepancy is 0.25 (near $t\approx 1$, where $\varepsilon=2$ is not small and there is a transient phase shift); it is 0.15 for $t\ge 5$ and 0.075 for $t\ge 20$. So the agreement is impressive even though ε is not small, as the book remarks: the slow algebraic decay $\sim t^{-1/2}$ of the amplitude is captured very accurately. ■

7.6.15 *(Pendulum)* $\ddot{x}+\sin x=0$. *(a) Using the method of Example 7.6.4, show the frequency of small oscillations of amplitude* $a\ll 1$ *is* $\omega\approx 1-\frac{1}{16}a^2$. *(b) Is this consistent with Exercise 6.7.4?*

Solution. *(a)* $\sin x\approx x-\frac{1}{6}x^3$, so $\ddot{x}+x-\frac{1}{6}x^3=0$, i.e. the Duffing equation of Example 7.6.4 with $h=x^3$ and $\varepsilon=-\frac{1}{6}$ (the cubic term is small because a is small). Example 7.6.4 gives $\omega=1+\frac{3}{8}\varepsilon a^2$, so

$$\omega\approx 1-\tfrac{3}{8}\cdot\tfrac{1}{6}a^2=1-\tfrac{1}{16}a^2.$$

(b) Yes. The exact period of the pendulum with amplitude a is $T=4K(\sin\frac{a}{2})=2\pi\big[1+\frac{1}{4}\sin^2\frac{a}{2}+\cdots\big]\approx 2\pi\big(1+\frac{1}{16}a^2\big)$, so $\omega=2\pi/T\approx 1-\frac{1}{16}a^2$, in agreement (the period increases with amplitude: the pendulum is a softening spring). ■

7.6.16 *(Amplitude of the van der Pol oscillator via Green's theorem) Assume the limit cycle is a circle of radius a and use $\oint_C \mathbf{v}\cdot\mathbf{n}\,\mathrm{d}l = \iint_A \nabla\cdot\mathbf{v}\,\mathrm{d}A$ with $\mathbf{v}=\dot{\mathbf{x}}=(\dot x,\dot y)$ to show $a\approx 2$.*

Solution. Write the van der Pol equation as $\dot x = y$, $\dot y = -x-\varepsilon(x^2-1)y$. The velocity field $\mathbf{v}=(\dot x,\dot y)$ is tangent to the closed orbit C, so $\mathbf{v}\cdot\mathbf{n}=0$ on C and the left side vanishes. The divergence is

$$\nabla\cdot\mathbf{v} = \frac{\partial y}{\partial x}+\frac{\partial}{\partial y}\big[-x-\varepsilon(x^2-1)y\big] = -\varepsilon(x^2-1).$$

So $0=-\varepsilon\iint_A (x^2-1)\,\mathrm{d}A$. For a disc of radius a,

$$\iint_A x^2\,\mathrm{d}A = \int_0^a\int_0^{2\pi}\rho^2\cos^2\vartheta\,\rho\,\mathrm{d}\vartheta\,\mathrm{d}\rho = \frac{\pi a^4}{4}, \qquad \iint_A 1\,\mathrm{d}A = \pi a^2.$$

Hence $\frac{\pi a^4}{4}=\pi a^2$, so $a^2=4$ and $a=2$. ■

7.6.17 *(Playing on a swing) $\ddot x+(1+\varepsilon\gamma+\varepsilon\cos 2t)\sin x=0$. (a) For small x, show the averaged equations are $r'=\frac14 r\sin 2\phi$, $\phi'=\frac12(\gamma+\frac12\cos 2\phi)$. (b) Show $r=0$ is unstable to exponentially growing oscillations if $|\gamma|<\gamma_c$. (c) Growth rate k. (d) Behaviour if $|\gamma|>\gamma_c$. (e) Physical interpretation.*

Solution. *(a)* For small x: $\ddot x + x + \varepsilon(\gamma+\cos 2t)x=0$, so $h=(\gamma+\cos 2t)\,x=(\gamma+\cos 2t)\,r\cos\theta$, where now $t=\theta-\phi$ and ϕ is held constant in the fast average. Then

$$r' = r\,\langle(\gamma+\cos 2t)\cos\theta\sin\theta\rangle = r\Big[\gamma\,\langle\cos\theta\sin\theta\rangle+\langle\cos(2\theta-2\phi)\cos\theta\sin\theta\rangle\Big].$$

The first average is 0. For the second, $\cos 2t\cos\theta\sin\theta=\frac12\cos(2\theta-2\phi)\sin 2\theta=\frac12(\cos 2\theta\cos 2\phi+\sin 2\theta\sin 2\phi)\sin 2\theta$, whose average is $\frac12\sin 2\phi\cdot\frac12=\frac14\sin 2\phi$. So $r'=\frac14 r\sin 2\phi$. Next,

$$r\phi' = r\,\big\langle(\gamma+\cos(2\theta-2\phi))\cos^2\theta\big\rangle = r\Big[\frac{\gamma}{2}+\big\langle\cos(2\theta-2\phi)\tfrac{1+\cos 2\theta}{2}\big\rangle\Big] = r\Big[\frac{\gamma}{2}+\frac14\cos 2\phi\Big],$$

so $\phi'=\frac{\gamma}{2}+\frac14\cos 2\phi=\frac12\big(\gamma+\frac12\cos 2\phi\big)$.

(b) For r near 0 the equation for ϕ does not involve r, so ϕ relaxes to an equilibrium: $\phi'=0\Rightarrow\cos 2\phi^*=-2\gamma$, which requires $|\gamma|\le\frac12$. The equilibrium is stable when $\mathrm{d}\phi'/\mathrm{d}\phi=-\frac12\sin 2\phi^*<0$, i.e. $\sin 2\phi^*=+\sqrt{1-4\gamma^2}$. Then

$$r' = \tfrac14\sqrt{1-4\gamma^2}\;r,$$

so $r(T)=r_0e^{kT}$ with $k>0$ whenever $|\gamma|<\gamma_c=\frac12$.

(c) $k=\frac14\sqrt{1-4\gamma^2}$ for $|\gamma|<\frac12$ (growth rate in the slow time $T=\varepsilon t$).

(d) For $|\gamma|>\frac12$ the right side of the ϕ equation never vanishes, so ϕ rotates steadily, and r goes up and down. Dividing the equations, $\mathrm{d}r/\mathrm{d}\phi=\dfrac{\frac14 r\sin 2\phi}{\frac12\gamma+\frac14\cos 2\phi}$, so $\mathrm{d}\ln r=\dfrac{\sin 2\phi\,\mathrm{d}\phi}{2\gamma+\cos 2\phi}$ and

$$r^2\,\big|2\gamma+\cos 2\phi\big| = \text{const}.$$

As ϕ is periodic, r is periodic in T and bounded: the origin is a (neutral) centre of the averaged equations, with no net growth or decay.

(e) This is *parametric resonance*. The pumping $\cos 2t$ is at (about) twice the swing's natural frequency. If the mismatch γ is small, $|\gamma|<\frac12$, any tiny initial motion grows exponentially, so the child can get the swing going from near rest just by pumping her legs, no push needed; if the pumping frequency is detuned more ($|\gamma|>\frac12$), the amplitude only oscillates and she needs a push to

get started. (As the amplitude grows, the small-angle approximation breaks down; the nonlinearity of $\sin x$ changes the frequency and limits the growth.) ■

7.6.18 *(Mathieu equation and a super-slow time scale)* $\ddot{x} + (a + \varepsilon\cos t)x = 0$, $a \approx 1$. *Using two-timing with* $T = \varepsilon^2 t$, *show that the solution becomes unbounded if* $1 - \frac{1}{12}\varepsilon^2 + O(\varepsilon^4) \le a \le 1 + \frac{5}{12}\varepsilon^2 + O(\varepsilon^4)$.

Solution. Let $\tau = t$, $T = \varepsilon^2 t$, $x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \cdots$ and $a = 1 + \varepsilon a_1 + \varepsilon^2 a_2 + \cdots$. Then $\ddot{x} = x_{\tau\tau} + 2\varepsilon^2 x_{\tau T} + O(\varepsilon^4)$.

$O(1)$: $x_{0\tau\tau} + x_0 = 0$, so $x_0 = A(T)\cos\tau + B(T)\sin\tau$.

$O(\varepsilon)$: $x_{1\tau\tau} + x_1 = -(a_1 + \cos\tau)x_0$. The terms $-a_1 A\cos\tau - a_1 B\sin\tau$ are resonant, so $a_1 = 0$. With $\cos^2\tau = \frac{1}{2}(1 + \cos 2\tau)$, $\sin\tau\cos\tau = \frac{1}{2}\sin 2\tau$:

$$x_{1\tau\tau} + x_1 = -\tfrac{A}{2} - \tfrac{A}{2}\cos 2\tau - \tfrac{B}{2}\sin 2\tau \implies x_1 = -\tfrac{A}{2} + \tfrac{A}{6}\cos 2\tau + \tfrac{B}{6}\sin 2\tau$$

(a particular solution; homogeneous parts are absorbed in x_0).

$O(\varepsilon^2)$: $x_{2\tau\tau} + x_2 = -2x_{0\tau T} - a_2 x_0 - \cos\tau\, x_1$. Here $-2x_{0\tau T} = 2A'\sin\tau - 2B'\cos\tau$ and, with $\cos\tau\cos 2\tau = \frac{1}{2}(\cos\tau + \cos 3\tau)$, $\cos\tau\sin 2\tau = \frac{1}{2}(\sin 3\tau + \sin\tau)$,

$$-\cos\tau\, x_1 = \tfrac{A}{2}\cos\tau - \tfrac{A}{12}(\cos\tau + \cos 3\tau) - \tfrac{B}{12}(\sin 3\tau + \sin\tau) = \tfrac{5A}{12}\cos\tau - \tfrac{B}{12}\sin\tau + (\text{third harmonics}).$$

Setting the coefficients of $\cos\tau$ and $\sin\tau$ to zero:

$$-2B' - a_2 A + \tfrac{5}{12}A = 0, \qquad 2A' - a_2 B - \tfrac{1}{12}B = 0,$$

i.e. $A' = \frac{1}{2}\big(a_2 + \frac{1}{12}\big)B$, $B' = \frac{1}{2}\big(\frac{5}{12} - a_2\big)A$. Hence

$$A'' = \tfrac{1}{4}\big(a_2 + \tfrac{1}{12}\big)\big(\tfrac{5}{12} - a_2\big)A.$$

Solutions grow exponentially in T iff $(a_2 + \frac{1}{12})(\frac{5}{12} - a_2) > 0$, i.e. $-\frac{1}{12} < a_2 < \frac{5}{12}$; otherwise they oscillate and stay bounded. Since $a = 1 + \varepsilon^2 a_2 + O(\varepsilon^4)$ (using $a_1 = 0$), the solution is unbounded for $1 - \frac{1}{12}\varepsilon^2 + O(\varepsilon^4) \le a \le 1 + \frac{5}{12}\varepsilon^2 + O(\varepsilon^4)$ (the endpoints correspond to linear, not exponential, growth or to boundary cases). ■

7.6.19 *(Poincaré–Lindstedt)* $\ddot{x} + x + \varepsilon x^3 = 0$, $x(0) = a$, $\dot{x}(0) = 0$. *(a) With* $\tau = \omega t$ *show* $\omega^2 x'' + x + \varepsilon x^3 = 0$. *(b) Substitute* $x = x_0 + \varepsilon x_1 + \cdots$, $\omega = 1 + \varepsilon\omega_1 + \varepsilon^2\omega_2 + \cdots$ *and show* $O(1)$*:* $x_0'' + x_0 = 0$; $O(\varepsilon)$*:* $x_1'' + x_1 = -2\omega_1 x_0'' - x_0^3$. *(c) Initial conditions. (d) Solve for* x_0. *(e) Show the* $O(\varepsilon)$ *equation becomes* $x_1'' + x_1 = (2\omega_1 a - \frac{3}{4}a^3)\cos\tau - \frac{1}{4}a^3\cos 3\tau$, *hence* $\omega_1 = \frac{3}{8}a^2$. *(f) Solve for* x_1.

Solution. *(a)* With $\tau = \omega t$, $\mathrm{d}/\mathrm{d}t = \omega\,\mathrm{d}/\mathrm{d}\tau$, so $\ddot{x} = \omega^2 x''$ and the equation becomes $\omega^2 x'' + x + \varepsilon x^3 = 0$ (primes: $\mathrm{d}/\mathrm{d}\tau$); the solution should be 2π-periodic in τ.

(b) $\omega^2 = 1 + 2\varepsilon\omega_1 + \varepsilon^2(\omega_1^2 + 2\omega_2) + \cdots$. Substituting and collecting:

$$O(1): x_0'' + x_0 = 0,$$
$$O(\varepsilon): x_1'' + 2\omega_1 x_0'' + x_1 + x_0^3 = 0 \implies x_1'' + x_1 = -2\omega_1 x_0'' - x_0^3.$$

(c) $x(\tau = 0) = a$ and $\omega x'(0) = 0 \Rightarrow x'(0) = 0$. Expanding, $x_0(0) + \varepsilon x_1(0) + \cdots = a$ for all ε, so $x_0(0) = a$ and $x_k(0) = 0$ for $k \ge 1$; likewise $x_0'(0) = 0$, $x_k'(0) = 0$ for $k \ge 1$.

(d) $x_0 = a\cos\tau$.

(e) $x_0'' = -a\cos\tau$ and $x_0^3 = a^3\cos^3\tau = \frac{a^3}{4}(3\cos\tau + \cos 3\tau)$. Hence

$$x_1'' + x_1 = 2\omega_1 a\cos\tau - \tfrac{3}{4}a^3\cos\tau - \tfrac{1}{4}a^3\cos 3\tau = \big(2\omega_1 a - \tfrac{3}{4}a^3\big)\cos\tau - \tfrac{1}{4}a^3\cos 3\tau.$$

The $\cos\tau$ term is resonant and would produce a secular term $\tau\sin\tau$; to avoid it, $2\omega_1 a = \frac{3}{4}a^3$, i.e. $\omega_1 = \frac{3}{8}a^2$.

(f) Now $x_1'' + x_1 = -\frac{1}{4}a^3\cos 3\tau$. A particular solution is $C\cos 3\tau$ with $(-9+1)C = -\frac{1}{4}a^3$, so $C = \frac{a^3}{32}$. Adding $D\cos\tau + E\sin\tau$ and imposing $x_1(0) = 0$ ($D = -\frac{a^3}{32}$) and $x_1'(0) = 0$ ($E = 0$):

$$x_1 = \tfrac{a^3}{32}\,(\cos 3\tau - \cos\tau).$$

So $x(t) \approx a\cos\omega t + \varepsilon\dfrac{a^3}{32}(\cos 3\omega t - \cos\omega t)$, $\omega = 1 + \frac{3}{8}\varepsilon a^2 + O(\varepsilon^2)$, uniformly valid for all t. ■

7.6.20 *Show that regular perturbation theory on 7.6.19 gives $x = a\cos t + \varepsilon a^3\big[-\frac{3}{8}t\sin t + \frac{1}{32}(\cos 3t - \cos t)\big] + O(\varepsilon^2)$. Why is this inferior?*

Solution. $O(1)$: $x_0 = a\cos t$. $O(\varepsilon)$: $\ddot{x}_1 + x_1 = -x_0^3 = -\frac{a^3}{4}(3\cos t + \cos 3t)$, $x_1(0) = \dot{x}_1(0) = 0$. For the resonant forcing $-\frac{3}{4}a^3\cos t$ use that $\ddot{y} + y = \cos t$ has particular solution $\frac{t}{2}\sin t$, giving $-\frac{3}{8}a^3 t\sin t$; for $-\frac{1}{4}a^3\cos 3t$ the particular solution is $\frac{a^3}{32}\cos 3t$ ($-8C = -\frac{1}{4}a^3$). So $x_1 = -\frac{3}{8}a^3 t\sin t + \frac{a^3}{32}\cos 3t + D\cos t + E\sin t$. The initial data $x_1(0) = 0$ and $\dot{x}_1(0) = 0$ give $D = -\frac{a^3}{32}$, $E = 0$. Therefore

$$x = a\cos t + \varepsilon a^3\Big[-\tfrac{3}{8}t\sin t + \tfrac{1}{32}(\cos 3t - \cos t)\Big] + O(\varepsilon^2).$$

This solution is inferior because of the secular term $-\frac{3}{8}\varepsilon a^3 t\sin t$, which grows without bound, whereas the true solution is periodic with bounded amplitude a (conservative system). The approximation is valid only for $t \ll 1/\varepsilon$. It is just the first terms of the expansion of $a\cos[(1+\frac{3}{8}\varepsilon a^2)t] = a\cos t - \frac{3}{8}\varepsilon a^3 t\sin t + \cdots$: the secular term is a frequency shift expanded in ε. The Poincaré–Lindstedt solution of 7.6.19 builds the frequency shift into $\cos\omega t$ and stays accurate for all t. ■

7.6.21 *Using Poincaré–Lindstedt, show the van der Pol limit cycle has frequency $\omega = 1 - \frac{1}{16}\varepsilon^2 + O(\varepsilon^3)$.*

Solution. With $\tau = \omega t$: $\omega^2 x'' + \varepsilon\omega(x^2-1)x' + x = 0$. Let $x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \cdots$, $\omega = 1 + \varepsilon\omega_1 + \varepsilon^2\omega_2 + \cdots$, with $x(0) = a$ (the amplitude, to be found) and $x'(0) = 0$ (choose the phase so that $\dot{x} = 0$ at $t = 0$; the limit cycle crosses $\dot{x} = 0$). Collect orders:

$$\begin{aligned} O(1) &: x_0'' + x_0 = 0,\\ O(\varepsilon) &: x_1'' + x_1 = -2\omega_1 x_0'' - (x_0^2 - 1)x_0',\\ O(\varepsilon^2) &: x_2'' + x_2 = -2\omega_1 x_1'' - (\omega_1^2 + 2\omega_2)x_0'' - \omega_1(x_0^2-1)x_0' - (x_0^2-1)x_1' - 2x_0 x_1 x_0'. \end{aligned}$$

$x_0 = a\cos\tau$, $x_0' = -a\sin\tau$, $x_0'' = -a\cos\tau$. The $O(\varepsilon)$ right side is

$$2\omega_1 a\cos\tau + \big(a^3\cos^2\tau - a\big)\sin\tau = 2\omega_1 a\cos\tau + \big(\tfrac{a^3}{4} - a\big)\sin\tau + \tfrac{a^3}{4}\sin 3\tau.$$

No secular terms: $\omega_1 = 0$ and $\frac{a^3}{4} - a = 0$, so $a = 2$. Then $x_1'' + x_1 = 2\sin 3\tau$, giving $x_1 = -\frac{1}{4}\sin 3\tau + D\cos\tau + E\sin\tau$; $x_1'(0) = 0$ gives $E = \frac{3}{4}$. (D is an $O(\varepsilon)$ correction to the amplitude, to be fixed at the next order.) So $x_1 = \frac{3}{4}\sin\tau - \frac{1}{4}\sin 3\tau + D\cos\tau = \sin^3\tau + D\cos\tau$ and $x_1' = 3\sin^2\tau\cos\tau - D\sin\tau$.

At $O(\varepsilon^2)$ (with $\omega_1 = 0$, $a = 2$, $x_0 = 2\cos\tau$) the right side is

$$4\omega_2\cos\tau - (1 + 2\cos 2\tau)\,x_1' + 8\cos\tau\sin\tau\,x_1 \qquad (\text{since } x_0^2 - 1 = 4\cos^2\tau - 1 = 1 + 2\cos 2\tau).$$

Using $\sin^2\tau\cos\tau = \frac{1}{4}(\cos\tau - \cos 3\tau)$ and product-to-sum formulas:

$$-(1 + 2\cos 2\tau)\,x_1' = -\tfrac{3}{4}\cos\tau + \tfrac{3}{4}\cos 5\tau + D\sin 3\tau,$$

$8\cos\tau\sin\tau\,x_1 = 8\sin^4\tau\cos\tau + 8D\cos^2\tau\sin\tau = \left[\cos\tau - \frac{3}{2}\cos 3\tau + \frac{1}{2}\cos 5\tau\right] + 2D\sin\tau + 2D\sin 3\tau.$

The resonant terms are therefore

$$\cos\tau:\ 4\omega_2 - \tfrac{3}{4} + 1 = 4\omega_2 + \tfrac{1}{4}, \qquad \sin\tau:\ 2D.$$

Setting both to zero: $D = 0$ and $\omega_2 = -\frac{1}{16}$. Hence

$$\omega = 1 - \tfrac{1}{16}\varepsilon^2 + O(\varepsilon^3),$$

and the limit cycle has amplitude $2 + O(\varepsilon^2)$ (since $D = 0$). ■

7.6.22 *(Asymmetric spring) Use Poincaré–Lindstedt to find the first few terms of the solution of $\ddot{x} + x + \varepsilon x^2 = 0$, $x(0) = a$, $\dot{x}(0) = 0$, and show the centre of oscillation is at $x \approx \mp\frac{1}{2}\varepsilon a^2$ (see the remark below on the sign).*

Solution. With $\tau = \omega t$, $\omega = 1 + \varepsilon\omega_1 + \varepsilon^2\omega_2 + \cdots$: $\omega^2 x'' + x + \varepsilon x^2 = 0$, $x_0 = a\cos\tau$ (with $x_k(0) = x_k'(0) = 0$, $k \geq 1$).

$O(\varepsilon)$: $x_1'' + x_1 = -2\omega_1 x_0'' - x_0^2 = 2\omega_1 a\cos\tau - \frac{a^2}{2}(1 + \cos 2\tau)$. Secular term: $\omega_1 = 0$. Then $x_1'' + x_1 = -\frac{a^2}{2} - \frac{a^2}{2}\cos 2\tau$, with particular solution $-\frac{a^2}{2} + \frac{a^2}{6}\cos 2\tau$. Adding $D\cos\tau + E\sin\tau$ with $x_1(0) = 0$, $x_1'(0) = 0$ gives $D = \frac{a^2}{3}$, $E = 0$:

$$x_1 = -\tfrac{a^2}{2} + \tfrac{a^2}{3}\cos\tau + \tfrac{a^2}{6}\cos 2\tau,$$

$$x \approx a\cos\omega t + \varepsilon a^2\left[-\tfrac{1}{2} + \tfrac{1}{3}\cos\omega t + \tfrac{1}{6}\cos 2\omega t\right].$$

$O(\varepsilon^2)$ *(frequency)*: $x_2'' + x_2 = -2\omega_2 x_0'' - 2x_0x_1 = 2\omega_2 a\cos\tau - 2x_0x_1$, and $x_0x_1 = a^3\left[-\frac{1}{2}\cos\tau + \frac{1}{12}(\cos\tau + \cos 3\tau) + \frac{1}{6}(1 + \cos 2\tau)\right]$ has $\cos\tau$ coefficient $-\frac{5}{12}a^3$. Removing the secular term: $2\omega_2 a + \frac{5}{6}a^3 = 0$, so $\omega_2 = -\frac{5}{12}a^2$ and

$$\omega = 1 - \tfrac{5}{12}\varepsilon^2 a^2 + O(\varepsilon^3).$$

Centre of oscillation. Averaging over a period, the oscillating terms $\cos\tau, \cos 2\tau$ drop out, so $\langle x\rangle = -\frac{1}{2}\varepsilon a^2 + O(\varepsilon^2)$. Check: averaging $\ddot{x} = -x - \varepsilon x^2$ over a period gives $0 = \langle x\rangle + \varepsilon\langle x^2\rangle$, so $\langle x\rangle \approx -\varepsilon\, a^2/2$.

Remark on the sign: the printed exercise says $x \approx \frac{1}{2}\varepsilon a^2$ without a minus sign. For $\ddot{x} + x + \varepsilon x^2 = 0$ with $\varepsilon > 0$, the restoring force $-x - \varepsilon x^2$ is stronger for $x < 0$ than for $x > 0$, so the oscillation is centred slightly to the *left* of the origin: $x \approx -\frac{1}{2}\varepsilon a^2$. The magnitude $\frac{1}{2}\varepsilon a^2$ agrees with the book; the center is on the negative side for $\varepsilon > 0$. ■

7.6.23 *Find the approximate relation between amplitude and frequency for periodic solutions of $\ddot{x} - \varepsilon x\dot{x} + x = 0$.*

Solution. At first order, averaging gives $h = -x\dot{x} = r^2\cos\theta\sin\theta$, so $r' = r^2\langle\cos\theta\sin^2\theta\rangle = 0$ and $r\phi' = r^2\langle\cos^2\theta\sin\theta\rangle = 0$: no amplitude or frequency change at $O(\varepsilon)$ (the system is reversible, so the origin is a centre and every nearby orbit is periodic). The amplitude–frequency relation therefore appears at $O(\varepsilon^2)$, and I use Poincaré–Lindstedt: $\tau = \omega t$, $\omega^2 x'' - \varepsilon\omega x x' + x = 0$, $x(0) = a$, $x'(0) = 0$, $\omega = 1 + \varepsilon\omega_1 + \varepsilon^2\omega_2$.

$O(1)$: $x_0 = a\cos\tau$.

$O(\varepsilon)$: $x_1'' + x_1 = -2\omega_1 x_0'' + x_0x_0' = 2\omega_1 a\cos\tau - \frac{a^2}{2}\sin 2\tau$. No secular term requires $\omega_1 = 0$. The particular solution is $\frac{a^2}{6}\sin 2\tau$ ($-3C = -\frac{a^2}{2}$); with $x_1(0) = 0$ and $x_1'(0) = 0$ (which gives $E = -\frac{a^2}{3}$):

$$x_1 = \tfrac{a^2}{6}\sin 2\tau - \tfrac{a^2}{3}\sin\tau, \qquad x_1' = \tfrac{a^2}{3}\cos 2\tau - \tfrac{a^2}{3}\cos\tau.$$

$O(\varepsilon^2)$: (with $\omega_1 = 0$) $x_2'' + x_2 = -2\omega_2 x_0'' + x_0 x_1' + x_1 x_0'$. The $\cos\tau$ coefficients: $-2\omega_2 x_0'' = 2\omega_2 a\cos\tau$; $x_0 x_1' = \frac{a^3}{3}(\cos\tau\cos 2\tau - \cos^2\tau)$ contributes $\frac{a^3}{6}$ (from $\cos\tau\cos 2\tau = \frac{1}{2}(\cos\tau + \cos 3\tau)$); $x_1 x_0' = -a\sin\tau\big(\frac{a^2}{6}\sin 2\tau - \frac{a^2}{3}\sin\tau\big)$ contributes $-\frac{a^3}{12}$ (from $\sin\tau\sin 2\tau = \frac{1}{2}(\cos\tau - \cos 3\tau)$). Setting the total $\cos\tau$ coefficient to zero:

$$2\omega_2 a + \tfrac{a^3}{6} - \tfrac{a^3}{12} = 0 \implies \omega_2 = -\tfrac{a^2}{24}.$$

Hence

$$\boxed{\omega \approx 1 - \frac{\varepsilon^2 a^2}{24} + O(\varepsilon^3)}$$

Check by direct numerical integration of the equation, measuring the period: $(\varepsilon, a) = (0.3, 1)$: $\omega = 0.99626$ vs. prediction 0.99625; $(0.5, 1)$: 0.98967 vs. 0.98958; $(0.5, 0.5)$: 0.99740 vs. 0.99740; $(0.3, 1.5)$: 0.99162 vs. 0.99156. ■

7.6.24 *(Computer algebra) Apply Poincaré–Lindstedt to $\ddot{x} + x - \varepsilon x^3 = 0$, $x(0) = a$, $\dot{x}(0) = 0$, and find ω up to and including the $O(\varepsilon^3)$ term.*

Solution. I carried out the method symbolically with SymPy (Python): with $\tau = \omega t$, $\omega = 1 + \sum_{k\ge 1}\varepsilon^k\omega_k$, $x = \sum_k \varepsilon^k x_k(\tau)$, at each order I removed the $\cos\tau$ resonant term from the forcing to determine ω_k, then solved $x_k'' + x_k =$ (remaining forcing) with $x_k(0) = x_k'(0) = 0$. The secular conditions came out as

$$\tfrac{a}{4}(3a^2 + 8\omega_1) = 0, \qquad \tfrac{a}{128}(21a^4 + 256\omega_2) = 0, \qquad \tfrac{a}{1024}(81a^6 + 2048\omega_3) = 0,$$

so

$$\omega_1 = -\tfrac{3}{8}a^2, \qquad \omega_2 = -\tfrac{21}{256}a^4, \qquad \omega_3 = -\tfrac{81}{2048}a^6,$$

$$\boxed{\omega = 1 - \tfrac{3}{8}\varepsilon a^2 - \tfrac{21}{256}\varepsilon^2 a^4 - \tfrac{81}{2048}\varepsilon^3 a^6 + O(\varepsilon^4)}$$

Check: the $O(\varepsilon)$ term is that of 7.6.19 with $\varepsilon \to -\varepsilon$ ($\omega_1 = -\frac{3}{8}a^2$). For the hardening Duffing oscillator $\ddot{x} + x + \varepsilon x^3 = 0$ the same computation gives $1 + \frac{3}{8}\varepsilon a^2 - \frac{21}{256}\varepsilon^2 a^4 + \frac{81}{2048}\varepsilon^3 a^6$ (the signs of the odd powers of ε flip), consistent with the exact period expansion of 7.6.13 at the orders we can compare ($c_2 = \frac{57\pi}{128}a^4$ corresponds to $\omega_2 = -\frac{21}{256}a^4$ via $\omega = 2\pi/T$). ■

7.6.25 *(The method of averaging) Consider $\ddot{x} + x + \varepsilon h(x, \dot{x}, t) = 0$ with $x = r\cos(t + \phi)$, $\dot{x} = -r\sin(t + \phi)$ (defining r, ϕ). (a) Show $\dot{r} = \varepsilon h\sin(t + \phi)$, $r\dot{\phi} = \varepsilon h\cos(t + \phi)$. (b) For the running average $\langle r\rangle(t) = \frac{1}{2\pi}\int_{t-\pi}^{t+\pi} r\,\mathrm{d}\tau$ show $\mathrm{d}\langle r\rangle/\mathrm{d}t = \langle\dot{r}\rangle$. (c) Show $\mathrm{d}\langle r\rangle/\mathrm{d}t = \varepsilon\langle h[r\cos(t + \phi), -r\sin(t + \phi), t]\sin(t + \phi)\rangle$. (d) Show $r = \bar{r} + O(\varepsilon)$, $\phi = \bar{\phi} + O(\varepsilon)$ and the averaged equations.*

Solution. Let $\theta = t + \phi$.

(a) Differentiating $x = r\cos\theta$ gives $\dot{x} = \dot{r}\cos\theta - r(1 + \dot{\phi})\sin\theta$. But by definition $\dot{x} = -r\sin\theta$, so

$$\dot{r}\cos\theta - r\dot{\phi}\sin\theta = 0. \tag{2}$$

Differentiating $\dot{x} = -r\sin\theta$ gives $\ddot{x} = -\dot{r}\sin\theta - r(1 + \dot{\phi})\cos\theta$. The equation $\ddot{x} + x + \varepsilon h = 0$ then reads $-\dot{r}\sin\theta - r\dot{\phi}\cos\theta + \varepsilon h = 0$, i.e.

$$\dot{r}\sin\theta + r\dot{\phi}\cos\theta = \varepsilon h. \tag{3}$$

Multiply (2) by $\cos\theta$, (3) by $\sin\theta$ and add: $\dot{r} = \varepsilon h\sin\theta$. Multiply (3) by $\cos\theta$, (2) by $\sin\theta$ and subtract: $r\dot{\phi} = \varepsilon h\cos\theta$.

(b) By the fundamental theorem of calculus, $\frac{\mathrm{d}}{\mathrm{d}t}\langle r\rangle = \frac{1}{2\pi}\big[r(t+\pi)-r(t-\pi)\big] = \frac{1}{2\pi}\int_{t-\pi}^{t+\pi}\dot r\,\mathrm{d}\tau = \langle \dot r\rangle$.

(c) Combining (a) and (b): $\dfrac{\mathrm{d}\langle r\rangle}{\mathrm{d}t} = \langle \dot r\rangle = \varepsilon\Big\langle h\big[r\cos(\tau+\phi), -r\sin(\tau+\phi), \tau\big]\sin(\tau+\phi)\Big\rangle$, where $\langle\cdot\rangle = \frac{1}{2\pi}\int_{t-\pi}^{t+\pi}\cdot\,\mathrm{d}\tau$ and $r = r(\tau)$, $\phi = \phi(\tau)$ inside.

(d) Since $\dot r = O(\varepsilon)$ and $r\dot\phi = O(\varepsilon)$, for $|\tau - t| \le \pi$ we have $r(\tau) = r(t) + O(\varepsilon)$ and $\phi(\tau) = \phi(t) + O(\varepsilon)$ (away from $r = 0$). Hence the running averages satisfy $\bar r(t) = r(t) + O(\varepsilon)$ and $\bar\phi(t) = \phi(t) + O(\varepsilon)$, i.e. $r(\tau) = \bar r(t) + O(\varepsilon)$, $\phi(\tau) = \bar\phi(t) + O(\varepsilon)$ on the averaging window. Since h is smooth, replacing r, ϕ inside the average by the constants $\bar r, \bar\phi$ changes the integrand by $O(\varepsilon)$ and the whole right side (which carries a factor ε) by $O(\varepsilon^2)$:

$$\frac{\mathrm{d}\bar r}{\mathrm{d}t} = \varepsilon\Big\langle h\big[\bar r\cos(\tau+\bar\phi), -\bar r\sin(\tau+\bar\phi), \tau\big]\sin(\tau+\bar\phi)\Big\rangle + O(\varepsilon^2).$$

Similarly $\mathrm{d}\bar\phi/\mathrm{d}t = \big\langle\dot\phi\big\rangle = \varepsilon\,\langle h\cos\theta/r\rangle = \varepsilon\,\big\langle h\cos(\tau+\bar\phi)\big\rangle/\bar r + O(\varepsilon^2)$, so

$$\bar r\,\frac{\mathrm{d}\bar\phi}{\mathrm{d}t} = \varepsilon\Big\langle h\big[\bar r\cos(\tau+\bar\phi), -\bar r\sin(\tau+\bar\phi), \tau\big]\cos(\tau+\bar\phi)\Big\rangle + O(\varepsilon^2).$$

For $h = h(x, \dot x)$ independent of time, changing the integration variable to $\theta = \tau + \bar\phi$ (with $\bar\phi$ constant over the window) turns these into $\bar r' = \langle h\sin\theta\rangle$, $\bar r\bar\phi' = \langle h\cos\theta\rangle$ in the slow time $T = \varepsilon t$: the averaged equations (7.6.53). ■

7.6.26 *(Calibrating the method of averaging) Consider $\dot x = -\varepsilon x\sin^2 t$, $0 \le \varepsilon \ll 1$, $x(0) = x_0$. (a) Find the exact solution. (b) With $\bar x = \frac{1}{2\pi}\int_{t-\pi}^{t+\pi} x\,\mathrm{d}\tau$, show $x = \bar x + O(\varepsilon)$; use averaging to find an approximate equation for $\bar x$ and solve it. (c) Compare: how large is the error?*

Solution. (In the printed exercise the equation is first order, $\dot x = -\varepsilon x\sin^2 t$; only one initial condition is given.)

(a) Separating variables, $\mathrm{d}x/x = -\varepsilon\sin^2 t\,\mathrm{d}t = -\frac{\varepsilon}{2}(1-\cos 2t)\,\mathrm{d}t$, so

$$x(t) = x_0\exp\Big[-\tfrac{\varepsilon}{2}t + \tfrac{\varepsilon}{4}\sin 2t\Big].$$

(b) Since $\dot x = O(\varepsilon)$, $x(\tau) = x(t) + O(\varepsilon)$ for $|\tau - t| \le \pi$, so $\bar x = x + O(\varepsilon)$ (more explicitly, the factor $e^{(\varepsilon/4)\sin 2t} = 1 + O(\varepsilon)$). Averaging the equation: $\dfrac{\mathrm{d}\bar x}{\mathrm{d}t} = \langle\dot x\rangle = -\varepsilon\,\big\langle x\sin^2\tau\big\rangle = -\varepsilon\bar x\,\big\langle\sin^2\tau\big\rangle + O(\varepsilon^2) = -\frac{\varepsilon}{2}\bar x + O(\varepsilon^2)$. Dropping the $O(\varepsilon^2)$ term and using $\bar x(0) = x_0 + O(\varepsilon)$:

$$\bar x(t) \approx x_0 e^{-\varepsilon t/2}.$$

(c) The exact solution is $x(t) = x_0 e^{-\varepsilon t/2}\,e^{(\varepsilon/4)\sin 2t} = x_0 e^{-\varepsilon t/2}\big[1 + \frac{\varepsilon}{4}\sin 2t + O(\varepsilon^2)\big]$. The averaged solution therefore differs from the exact one by $x_0 e^{-\varepsilon t/2}\big[\frac{\varepsilon}{4}\sin 2t + O(\varepsilon^2)\big]$: an *oscillating* relative error of size at most $e^{\varepsilon/4} - 1 \approx \varepsilon/4$, which is $O(\varepsilon)$, uniformly in t (it does not grow). Averaging has thrown away only the small fast wiggle $\frac{\varepsilon}{4}\sin 2t$ in the exact solution, but captured the slow exponential decay at rate $\varepsilon/2$ exactly. ■